

## Basic OS Interview Questions

### 1. What is IPC? What are the different IPC mechanisms?

IPC (Interprocess Communication) is a mechanism that requires the use of resources like a memory that is shared between processes or threads. With IPC, OS allows different processes to communicate with each other. It is simply used for exchanging data between multiple threads in one or more programs or processes. In this mechanism, different processes can communicate with each other with the approval of the OS.

Different IPC Mechanisms:

- Pipes
- Message Queuing
- Semaphores
- Socket
- Shared Memory
- Signals

### 2. What's the main purpose of an OS? What are the different types of OS?

The main purpose of an OS is to execute user programs and make it easier for users to understand and interact with computers as well as run applications. It is specially designed to ensure that the computer system performs better by managing all computational activities. It also manages computer memory, processes, and operation of all hardware and software.

Types of OS:

- Batched OS (Example: Payroll System, Transactions Process, etc.)
- Multi-Programmed OS (Example: Windows O/S, UNIX O/S, etc.)
- Timesharing OS (Example: Multics, etc.)
- Distributed OS (LOCUS, etc.)
- Real-Time OS (PSOS, VRTX, etc.)

### 3. What are the benefits of a multiprocessor system?

A Multiprocessor system is a type of system that includes two or more CPUs. It involves the processing of different computer programs at the same time mostly by a computer system with two or more CPUs that are sharing single memory.

Benefits:

- Such systems are used widely nowadays to improve performance in systems that are running multiple programs concurrently.
- By increasing the number of processors, a greater number of tasks can be completed in unit time.
- One also gets a considerable increase in throughput and is cost-effective also as all processors share the same resources.
- It simply improves the reliability of the computer system.

### 4. What is RAID structure in OS? What are the different levels of RAID configuration?

RAID (Redundant Arrays of Independent Disks) is a method used to store data on Multiple hard disks therefore it is considered as data storage virtualization technology that combines multiple hard disks. It simply balances data protection, system performance, storage space, etc. It is used to improve the overall performance and reliability of data storage. It also increases the storage capacity of the system and its main purpose is to achieve data redundancy to reduce data loss.

Different levels of RAID

Nowadays, RAID is available in various schemes or RAID level as given below:

|         |       |                                      |
|---------|-------|--------------------------------------|
| LEVEL 6 | ————→ | P+Q Redundancy                       |
| LEVEL 5 | ————→ | Block interleaved distributed Parity |
| LEVEL 4 | ————→ | Block interleaved parity             |
| LEVEL 3 | ————→ | Bit interleaved parity               |
| LEVEL 2 | ————→ | Memory style error correcting code   |
| LEVEL 1 | ————→ | Mirroring & Duplexing                |
| LEVEL 0 | ————→ | Non-Redundant Striping               |



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LEVEL 1

LEVEL 0

Mirroring & Duplexing

Non-Redundant Striping



- RAID 0 - Non-redundant striping: This level is used to increase the performance of the server.
- RAID 1 - Mirroring and duplexing: This level is also known as disk mirroring and is considered the simplest way to implement fault tolerance.
- RAID 2 - Memory-style error-correcting codes: This level generally uses dedicated hamming code parity I.e., a liner form of error correction code.
- RAID 3 - Bit-interleaved Parity: This level requires a dedicated parity drive to store parity information.
- RAID 4 - Block-interleaved Parity: This level is similar to RAID 5 but the only difference is that this level confines all parity data to a single drive.
- RAID 5 - Block-interleaved distributed Parity: This level provides far better performance than disk mirroring and fault tolerance.
- RAID 6 - P+Q Redundancy: This level generally provides fault tolerance for two drive failures.

## 5. What is GUI?

GUI (Graphical User Interface) is basically a type of user interface that allows users to use graphics to interact with OS. GUI is created because it is more user-friendly, less complex, and easier to understand rather than a command-line interface. Its main goal is to increase efficiency and ease of use. Instead of having to memorize commands, users can just click on a button to simply execute the procedure. Examples of GUI include Microsoft Windows, macOS, Apple's iOS, etc.

## 6. What is a Pipe and when it is used?

The pipe is generally a connection among two or more processes that are interrelated to each other. It is a mechanism that is used for inter-process communication using message passing. One can easily send information such as the output of one program process to another program process using a pipe. It can be used when two processes want to communicate one-way i.e., inter-process communication (IPC).

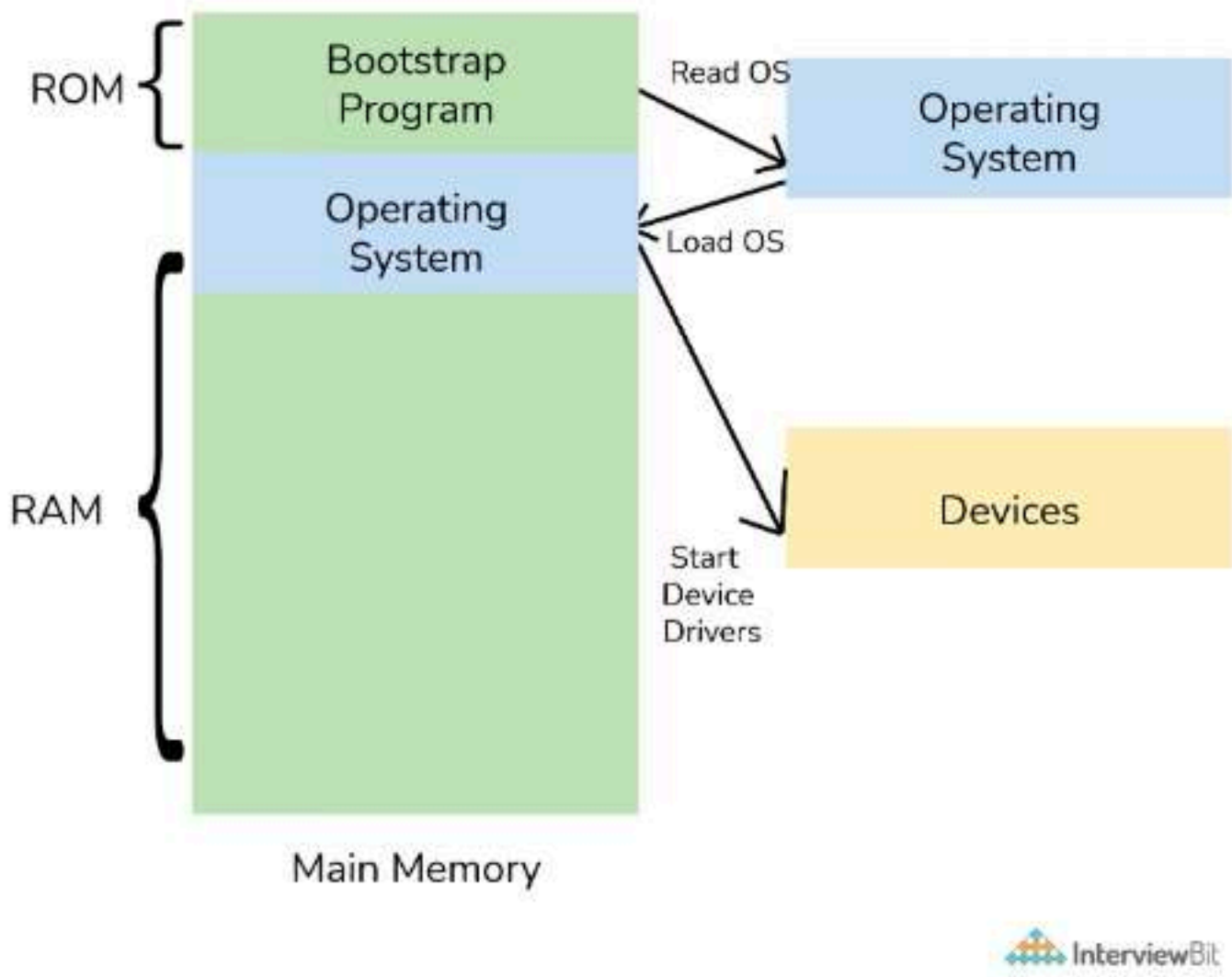
## 7. What are the different kinds of operations that are possible on semaphore?

There are basically two atomic operations that are possible:

- Wait()
- Signal()

## 8. What is a bootstrap program in OS?

It is generally a program that initializes OS during startup i.e., first code that is executed whenever computer system startups. OS is loaded through a bootstrapping process or program commonly known as booting. Overall OS only depends on the bootstrap program to perform and work correctly. It is fully stored in boot blocks at a fixed location on the disk. It also locates the kernel and loads it into the main memory after which the program starts its execution.



## 9. Explain demand paging?

Demand paging is a method that loads pages into memory on demand. This method is mostly used in virtual memory. In this, a page is only brought into memory when a location on that particular page is referenced during execution. The following steps are generally followed:

- Attempt to access the page.
- If the page is valid (in memory) then continue processing instructions as normal.
- If a page is invalid then a **page-fault trap** occurs.
- Check if the memory reference is a valid reference to a location on secondary memory. If not, the process is terminated (**illegal memory access**). Otherwise, we have to **page in** the required page.
- Schedule disk operation to read the desired page into main memory.
- Restart the instruction that was interrupted by the operating system trap.

## 10. What do you mean by RTOS?

Real Time Operating System (RTOS) is an operating system that is used for real-t

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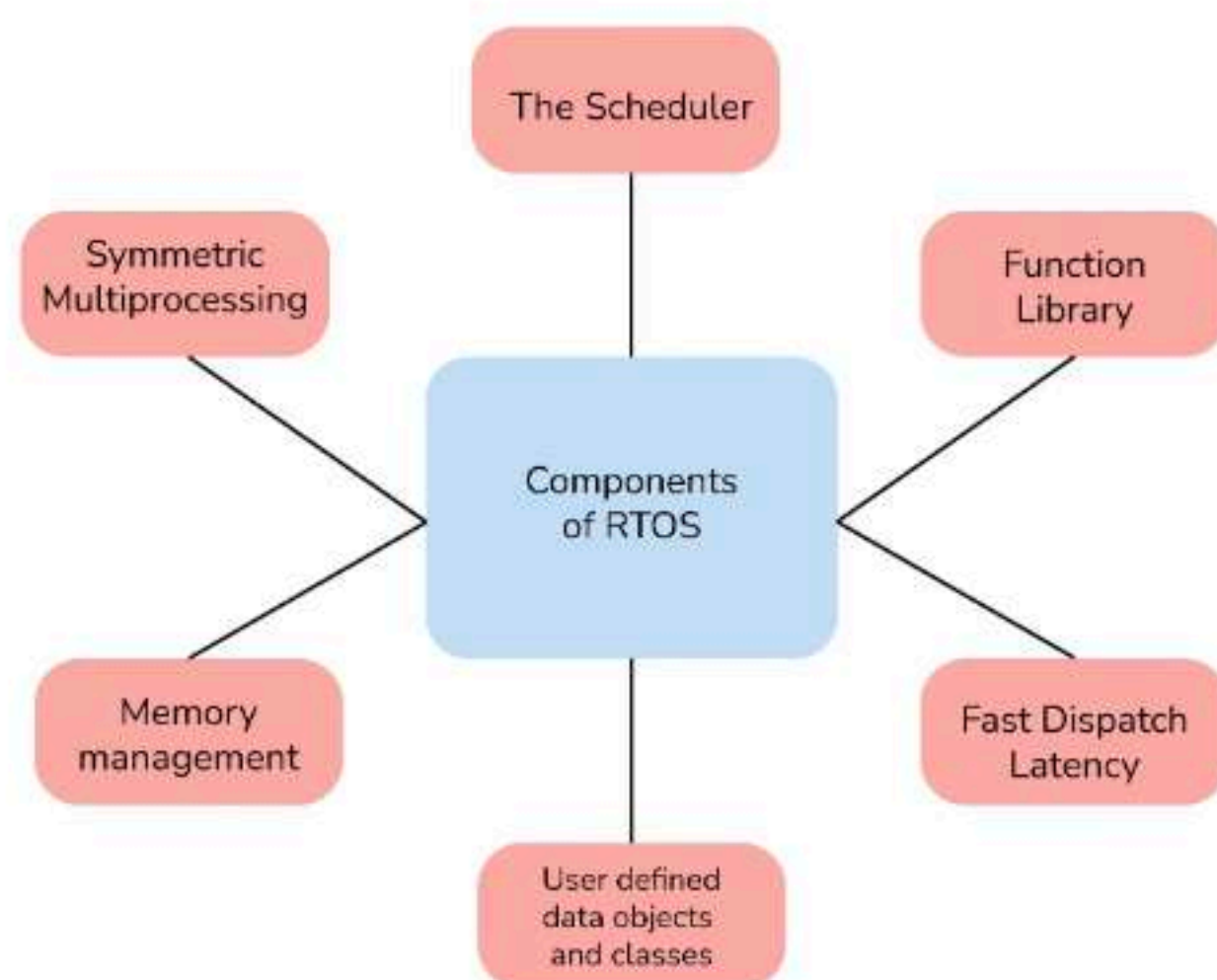
- Schedule disk operation to read the desired page into main memory.
- Restart the instruction that was interrupted by the operating system trap.

## 10. What do you mean by RTOS?

Real Time Operating System (RTOS) is an operating system that is used for real-time applications i.e., for those applications where data processing should be done in a fixed and small measure of time. It performs much better on tasks that are needed to be executed within a short time. It also takes care of execution, monitoring, and all-controlling processes. It also occupies less memory and consumes fewer resources.

Types of RTOS:

- Hard Real-Time
- Firm Real-Time
- Soft Real-Time



RTOS is used in Air traffic control systems, Anti-lock Brake Systems, and Heart pacemakers.

## 11. What do you mean by process synchronization?

Process synchronization is basically a way to coordinate processes that use shared resources or data. It is very much essential to ensure synchronized execution of cooperating processes so that will maintain data consistency. Its main purpose is to share resources without any interference using mutual exclusion. There are two types of process synchronization:

- Independent Process
- Cooperative Process

## 12. Why is the operating system important?

OS is the most essential and vital part of a computer without which it is considered useless. It enables an interface or acts like a link for interaction between computer software that is installed on OS and users. It also helps to communicate with hardware and also maintains balance among hardware and CPU. It also provides services to users and a platform for programs to run on. It performs all common tasks applications require.

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# Linux vs Windows

## All You Need to Know



**Priya Kadakia**  
Research Analyst

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## 13. What is different between main memory and secondary memory.

**Main memory:** Main memory in a computer is RAM (Random Access Memory). It is also known as primary memory or read-write memory or internal memory. The programs and data that the CPU requires during the execution of a program are stored in this memory.

**Secondary memory:** Secondary memory in a computer are storage devices that can store data and programs. It is also known as external memory or additional memory or backup memory or auxiliary memory. Such storage devices are capable of storing high-volume data. Storage devices can be hard drives, USB flash drives, CDs, etc.

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| Primary Memory                                                                                 | Secondary Memory                                                                                       |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Data can be directly accessed by the processing unit.                                          | Firstly, data is transferred to primary memory and after then routed to the processing unit.           |
| It can be both volatile and non-volatile in nature.                                            | It is non-volatile in nature.                                                                          |
| It is more costly than secondary memory.                                                       | It is more cost-effective or less costly than primary memory.                                          |
| It is temporary because data is stored temporarily.                                            | It is permanent because data is stored permanently.                                                    |
| In this memory, data can be lost whenever there is a power failure.                            | In this memory, data is stored permanently and therefore cannot be lost even in case of power failure. |
| It is much faster than secondary memory and saves data that is currently used by the computer. | It is slower as compared to primary memory and saves different kinds of data in different formats.     |
| It can be accessed by data.                                                                    | It can be accessed by I/O channels.                                                                    |

### 14. What do you mean by overlays in OS?

Overlays is basically a programming method that divides processes into pieces so that instructions that are important and need can be saved in memory. It does not need any type of support from the OS. It can run programs that are bigger in size than physical memory by only keeping only important data and instructions that can be needed at any given time.

### 15. Write top 10 examples of OS?

Some of the top OS's that are used mostly are given below:

- MS-Windows
- Ubuntu
- Mac OS
- Fedora
- Solaris
- Free BSD
- Chrome OS
- CentOS
- Debian
- Android

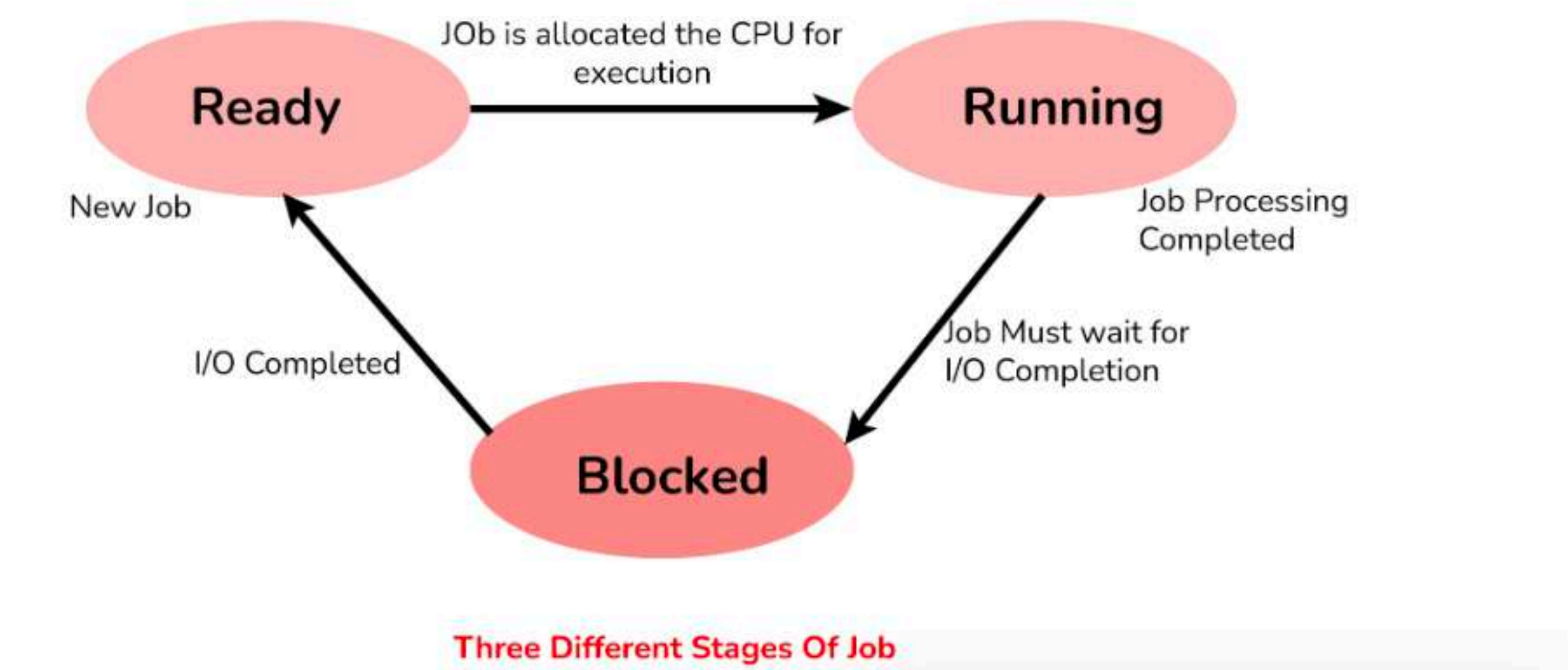
## Intermediate OS Interview Questions

### 16. What is thrashing in OS?

It is generally a situation where the CPU performs less productive work and more swapping or paging work. It spends more time swapping or paging activities rather than its execution. By evaluating the level of CPU utilization, a system can detect thrashing. It occurs when the process does not have enough pages due to which the page-fault rate is increased. It inhibits much application-level processing that causes computer performance to degrade or collapse.

### 17. What is the main objective of multiprogramming?

It refers to the ability to execute or perform more than one program on a single processor machine. This technique was introduced to overcome the problem of underutilization of CPU and main memory. In simple words, it is the coordination of execution of various programs simultaneously on a single processor (CPU). The main objective of multiprogramming is to have at least some processes running at all times. It simply improves the utilization of the CPU as it organizes many jobs where the CPU always has one to execute.

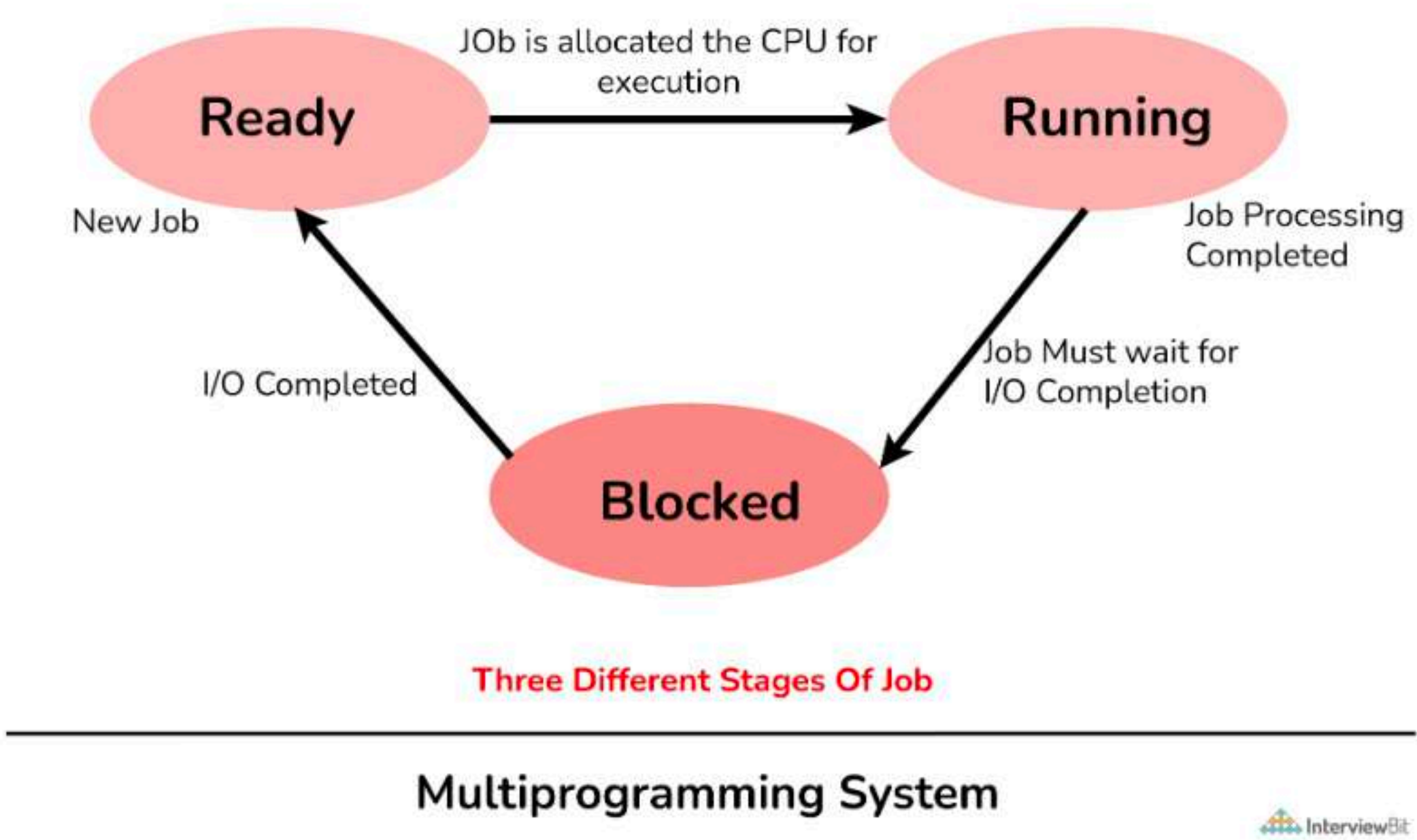




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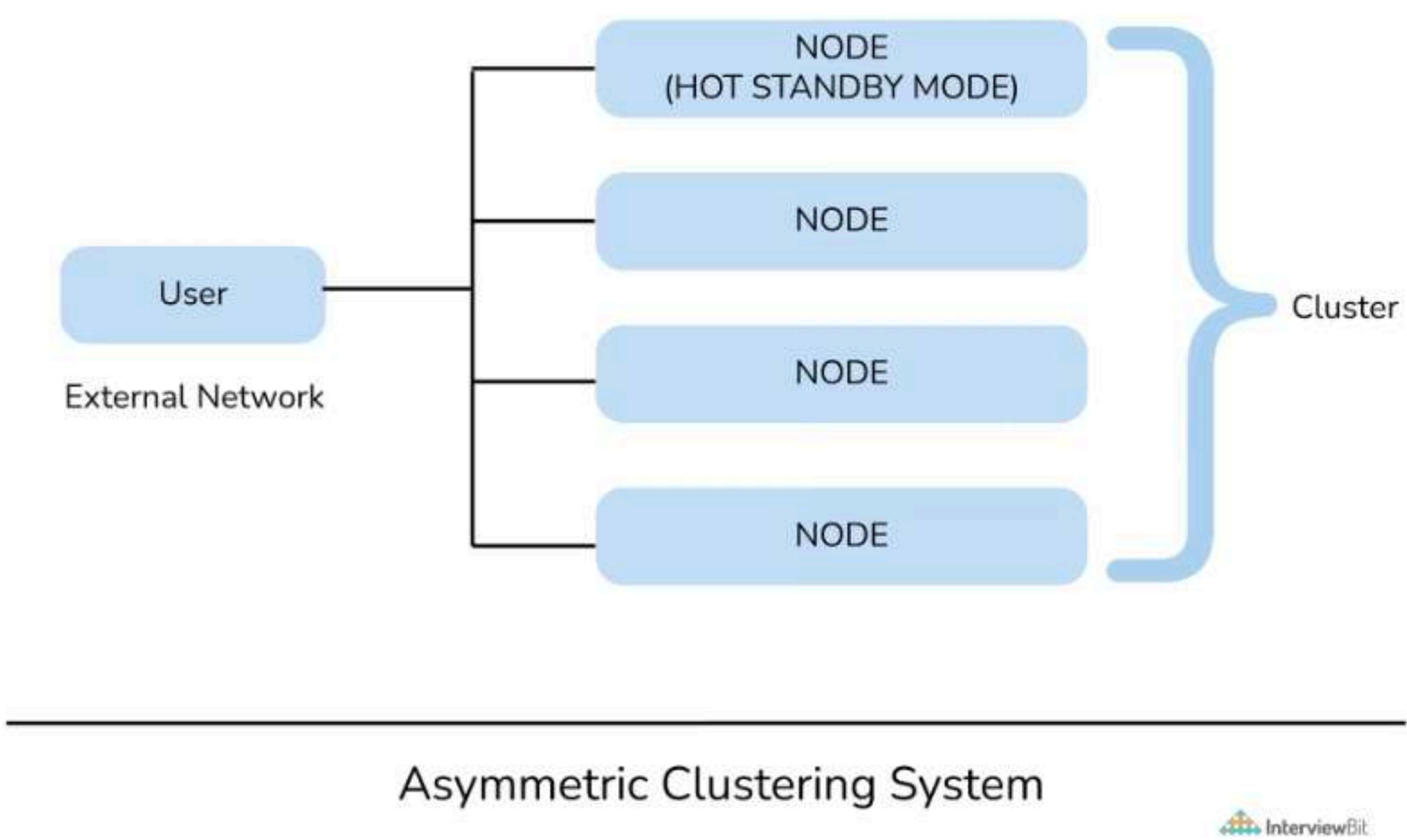
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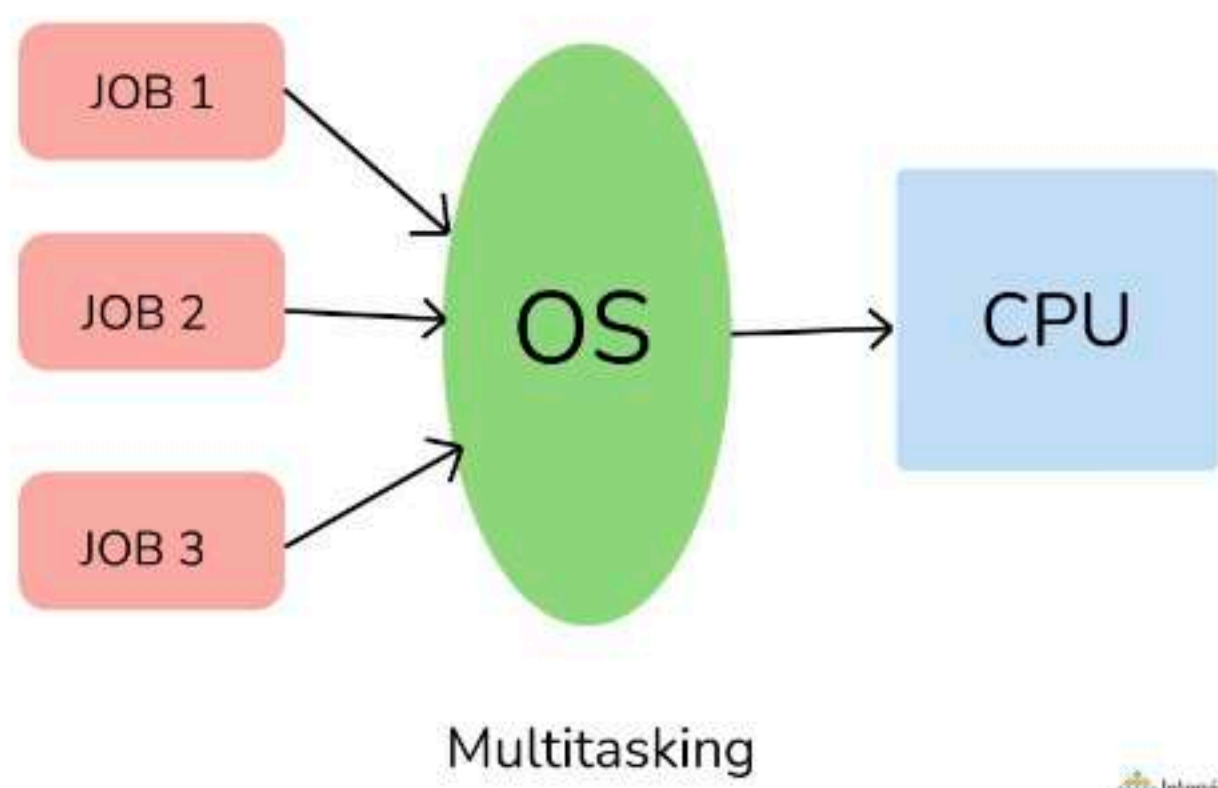
### 18. What do you mean by asymmetric clustering?

Asymmetric Clustering is generally a system in which one of the nodes among all nodes is in hot standby mode whereas the rest of all nodes run different applications. It simply uses whole or entire hardware resources therefore it is considered a more reliable system as compared to others.



### 19. What is the difference between multitasking and multiprocessing OS?

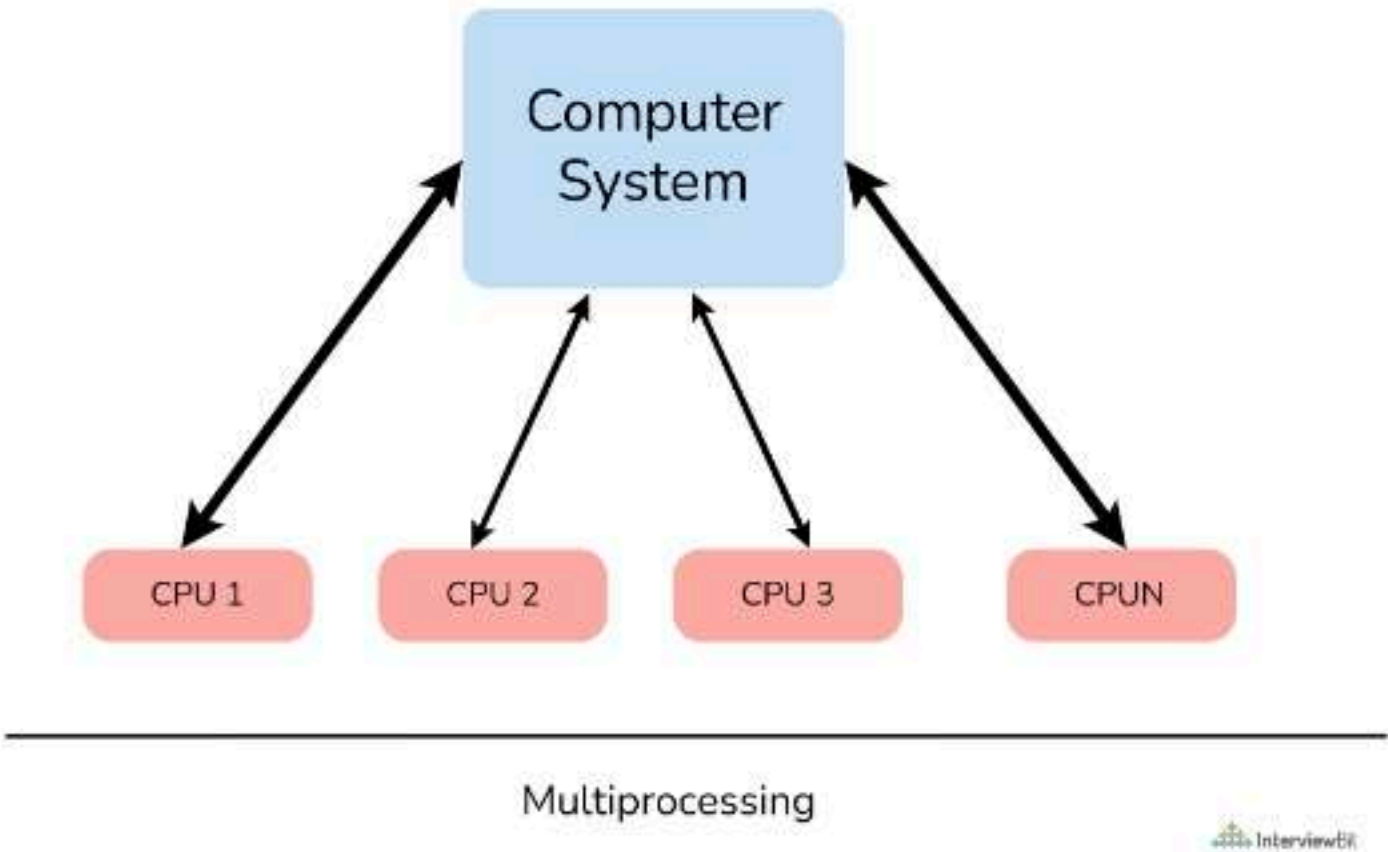
**Multitasking:** It is a system that allows more efficient use of computer hardware. This system works on more than one task at one time by rapidly switching between various tasks. These systems are also known as time-sharing systems.



**Multiprocessing:** It is a system that allows multiple or various processors in a computer to process two or more different portions of the same program simultaneously. It is used to complete more work in a shorter



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| Multitasking                                                           | Multiprocessing                                                       |
|------------------------------------------------------------------------|-----------------------------------------------------------------------|
| It performs more than one task at a time using a single processor.     | It performs more than one task at a time using multiple processors.   |
| In this, the number of CPUs is only one.                               | In this, the number of CPUs is more than one.                         |
| It is more economical.                                                 | It is less economical.                                                |
| It is less efficient than multiprocessing.                             | It is more efficient than multitasking.                               |
| It allows fast switching among various tasks.                          | It allows smooth processing of multiple tasks at once.                |
| It requires more time to execute tasks as compared to multiprocessing. | It requires less time for job processing as compared to multitasking. |

## 20. What do you mean by Sockets in OS?

The socket in OS is generally referred to as an endpoint for IPC (Interprocess Communication). Here, the endpoint is referred to as a combination of an IP address and port number. Sockets are used to make it easy for [software developers](#) to create network-enabled programs. It also allows communication or exchange of information between two different processes on the same or different machines. It is mostly used in client-server-based systems.

### Types of Sockets

There are basically four types of sockets as given below:

- Stream Sockets
- Datagram Sockets
- Sequenced Packet Sockets
- Raw Sockets

## 21. Explain zombie process?

Zombie process, referred to as a defunct process, is basically a process that is terminated or completed but the whole process control block is not cleaned up from the main memory because it still has an entry in the process table to report to its parent process. It does not consume any of the resources and is dead, but it still exists. It also shows that resources are held by process and are not free.

## 22. What do you mean by cascading termination?

Cascading termination is a process termination in which if the parent process is exiting or terminating then the children process will also get terminated. It does not allow the child to continue processing as its parent process terminates. It is generally initiated by OS.

## 23. What is starvation and aging in OS?

When we use Priority Scheduling or Shortest Job First Scheduling, Starvation can happen, This algorithm is mostly used in CPU schedulers

**Starvation:** It is generally a problem that usually occurs when a process has not been able to get the required resources it needs for progress with its execution for a long period of time. In this condition, low priority processes get blocked and only high priority processes proceed towards completion because of which low priority processes suffer from lack of resources.

**Aging:** It is a technique that is used to overcome the situation or problem of starvation that wait in the system for resources for a long period of time. It is considered the solution to the problem of starvation as it adds an aging factor to the priority of each and every request by increasing its priority over time.



OS.

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**Aging:** It is a technique that is used to overcome the situation or problem of starvation. It simply increases the priority of processes that wait in the system for resources for a long period of time. It is considered the best technique to resolve the problem of starvation as it adds an aging factor to the priority of each and every request by various processes for resources. It also ensures that low-level queue jobs or processes complete their execution.

### 24. What is the difference between paging and segmentation?

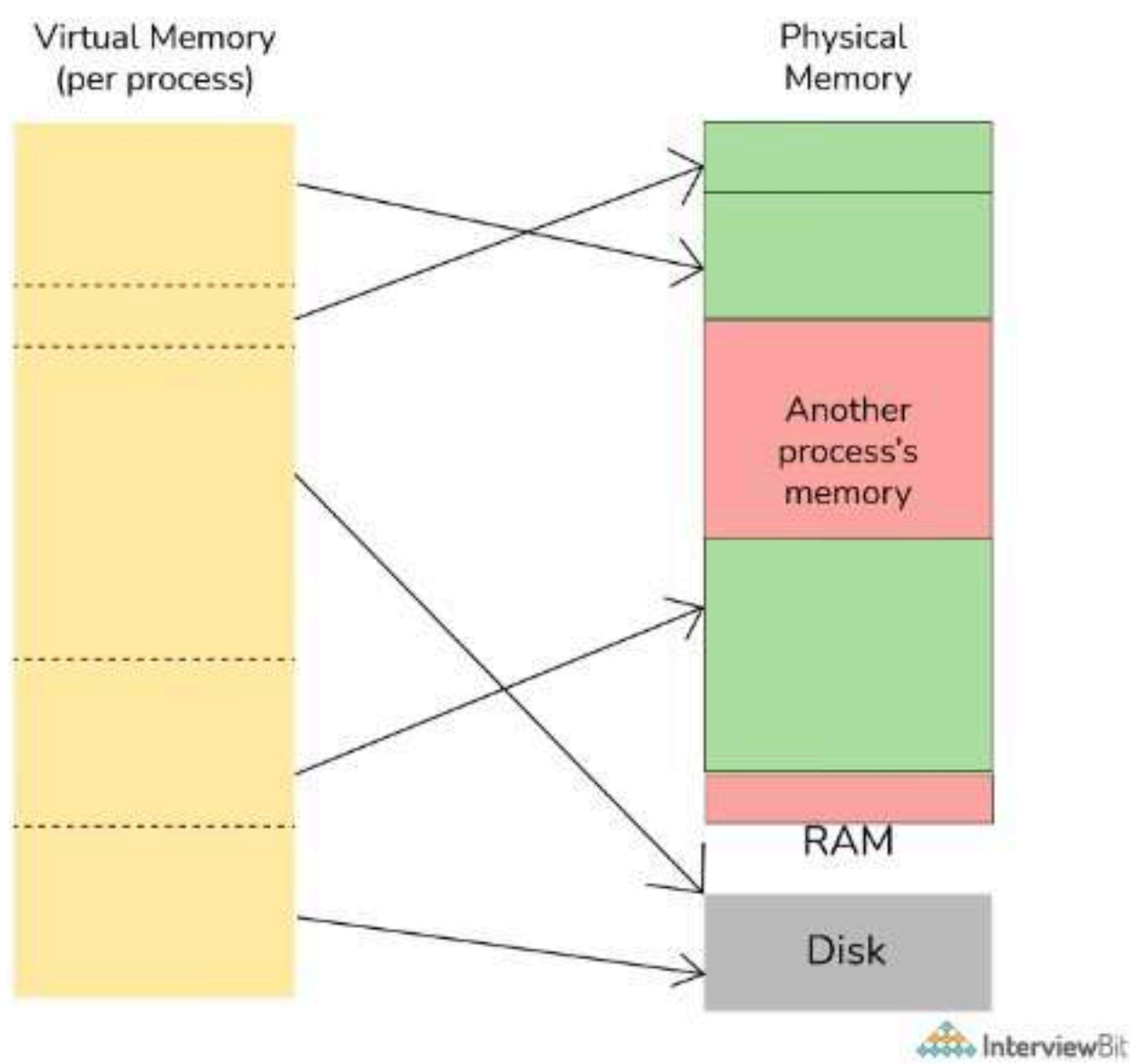
**Paging:** It is generally a memory management technique that allows OS to retrieve processes from secondary storage into main memory. It is a non-contiguous allocation technique that divides each process in the form of pages.

**Segmentation:** It is generally a memory management technique that divides processes into modules and parts of different sizes. These parts and modules are known as segments that can be allocated to process.

| Paging                                                                                | Segmentation                                                                                                        |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| It is invisible to a programmer.                                                      | It is visible to a programmer.                                                                                      |
| In this, the size of pages is fixed.                                                  | In this, the size of segments is not fixed.                                                                         |
| Procedures and data cannot be separated in paging.                                    | Procedures and data can be separated in segmentation.                                                               |
| It allows a cumulative total of virtual address spaces to cross physical main memory. | It allows all programs, data, and codes to break up into independent address spaces.                                |
| It is mostly available on CPUs and MMU chips.                                         | It is mostly available on Windows servers that may support backward compatibility, while Linux has limited support. |
| It is faster for memory access as compared to segmentation.                           | It is slower as compared to paging.                                                                                 |
| In this, OS needs to maintain a free frame.                                           | In this, OS needs to maintain a list of holes in the main memory.                                                   |
| In paging, the type of fragmentation is internal.                                     | In segmentation, the type of fragmentation is external.                                                             |
| The size of the page is determined by available memory.                               | The size of the page is determined by the user.                                                                     |

### 25. What is virtual memory?

It is a memory management technique feature of OS that creates the illusion to users of a very large (main) memory. It is simply space where a greater number of programs can be stored by themselves in the form of pages. It enables us to increase the use of physical memory by using a disk and also allows us to have memory protection. It can be managed in two common ways by OS i.e., [paging and segmentation](#). It acts as temporary storage that can be used along with RAM for computer processes.



### 26. What is thread in OS?

Thread is a path of execution that is composed of a program counter, thread id, stack, and set of registers within the process. It is a basic unit of CPU utilization that makes communication more effective and efficient, enables utilization of multiprocessor architectures to a greater scale and greater efficiency, and reduces the time required in context switching. It simply provides a way to improve and increase the performance of applications through parallelism. Threads are sometimes called **lightweight processes** because they have their own stack but can access shared data.

Multiple threads running in a process share: Address space, Heap, Static data, Code segments, File descriptors, Global variables, Child processes, Pending alarms, Signals, and signal handlers.

Each thread has its own: Program counter, Registers, Stack, and State.

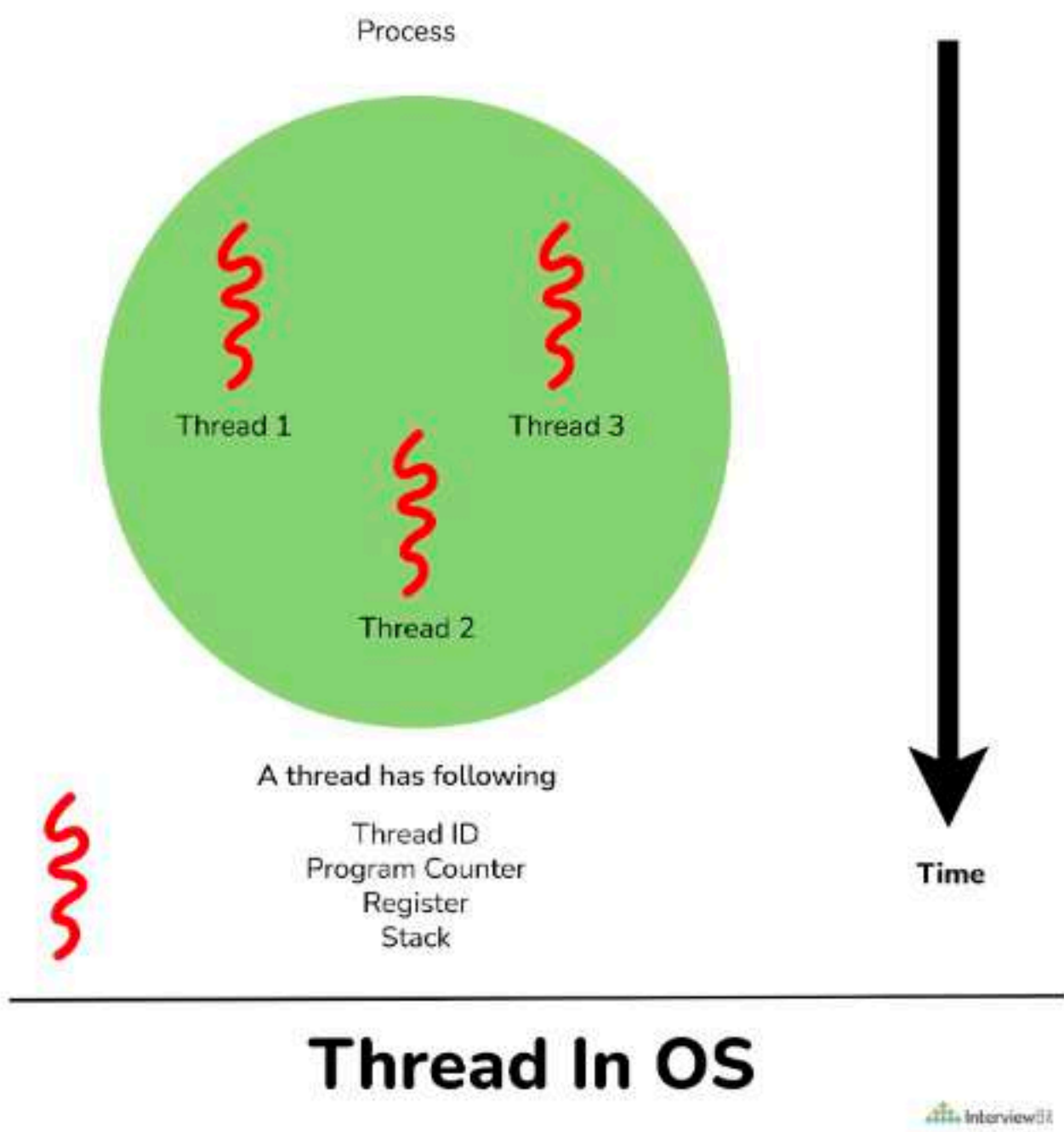


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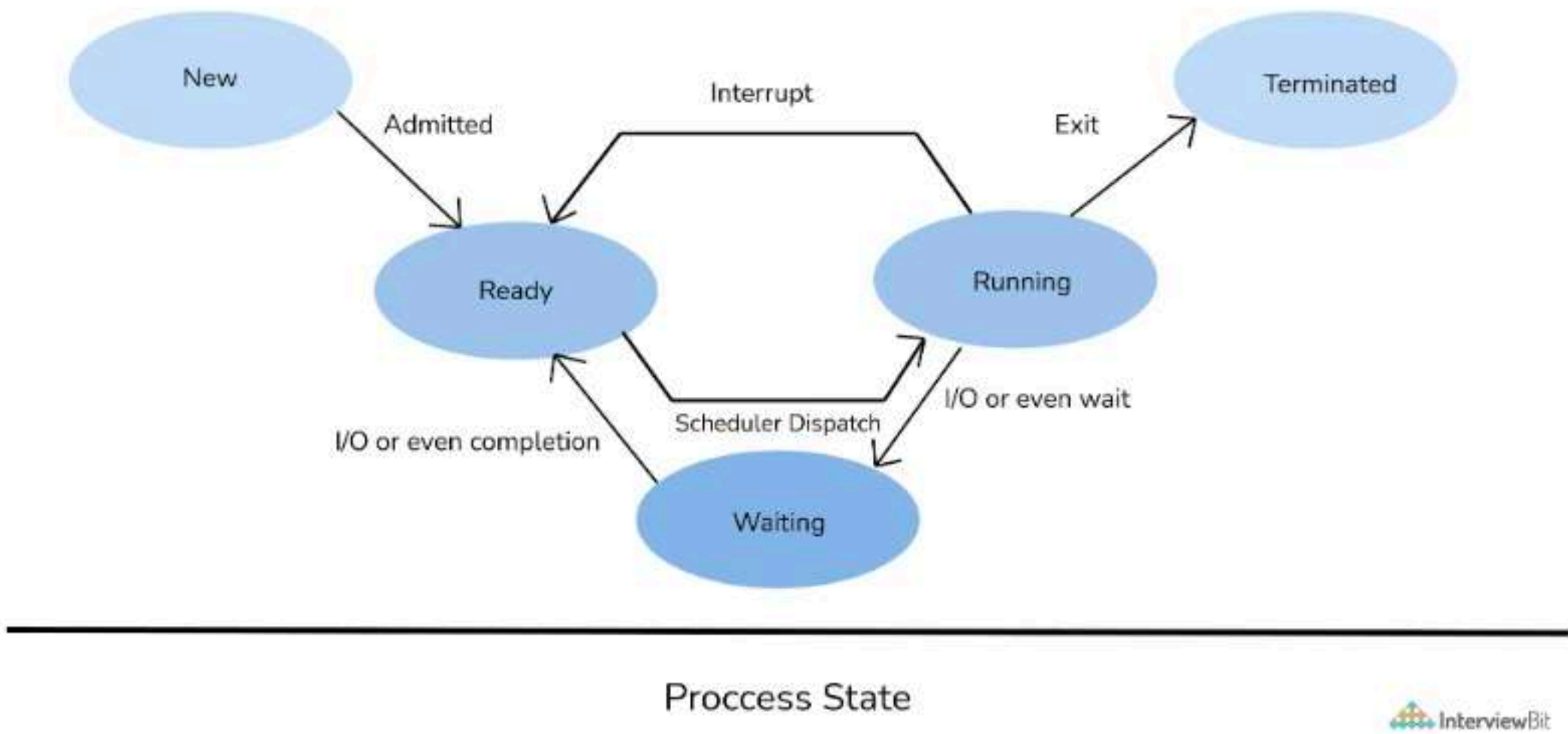
Each thread has its own: Program counter, Registers, Stack, and State.



## 27. What is a process? What are the different states of a process?

The process is basically a program that is currently under execution. The main function of an OS is to manage and handle all of these processes. When a program is loaded into the memory and it becomes a process, it can be divided into four sections — stack, heap, text, and data. There are two types of processes:

1. Operating System Processes
2. User Processes



### States of Process:

Different states of the process through which process goes are given below:

- **New State:** In this state, a process is just created.
- **Running:** In this state, the CPU starts working on the process's instructions.
- **Waiting:** In this state, the process cannot run because it just waits for some event to occur
- **Ready:** In this state, the process has all resources available that are required to run but it waits to get assigned to a processor because CPUs are not working currently on instructions passed by the process.
- **Terminate:** In this state, the process is completed I.e., the process has finished execution.

## 28. What do you mean by FCFS?

FCFS (First Come First Serve) is a type of OS scheduling algorithm that executes processes in the same order in which processes arrive. In simple words, the process that arrives first will be executed first. It is non-preemptive in nature. FCFS scheduling may cause the problem of starvation if the burst time of the first process is the longest among all the jobs. Burst time here means the time that is required in milliseconds by the process for its execution. It is also considered the easiest and simplest OS scheduling algorithm as compared to others. Implementation of FCFS is generally managed with help of the FIFO (First In First Out) queue.

## 29. What is Reentrancy?



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29. What is Reentrancy?

Reentrant is simply a function in which various clients can use and shares a single copy of a program during a similar period. This concept is generally associated with OS code and does not deal with concurrency. It has two major functions:

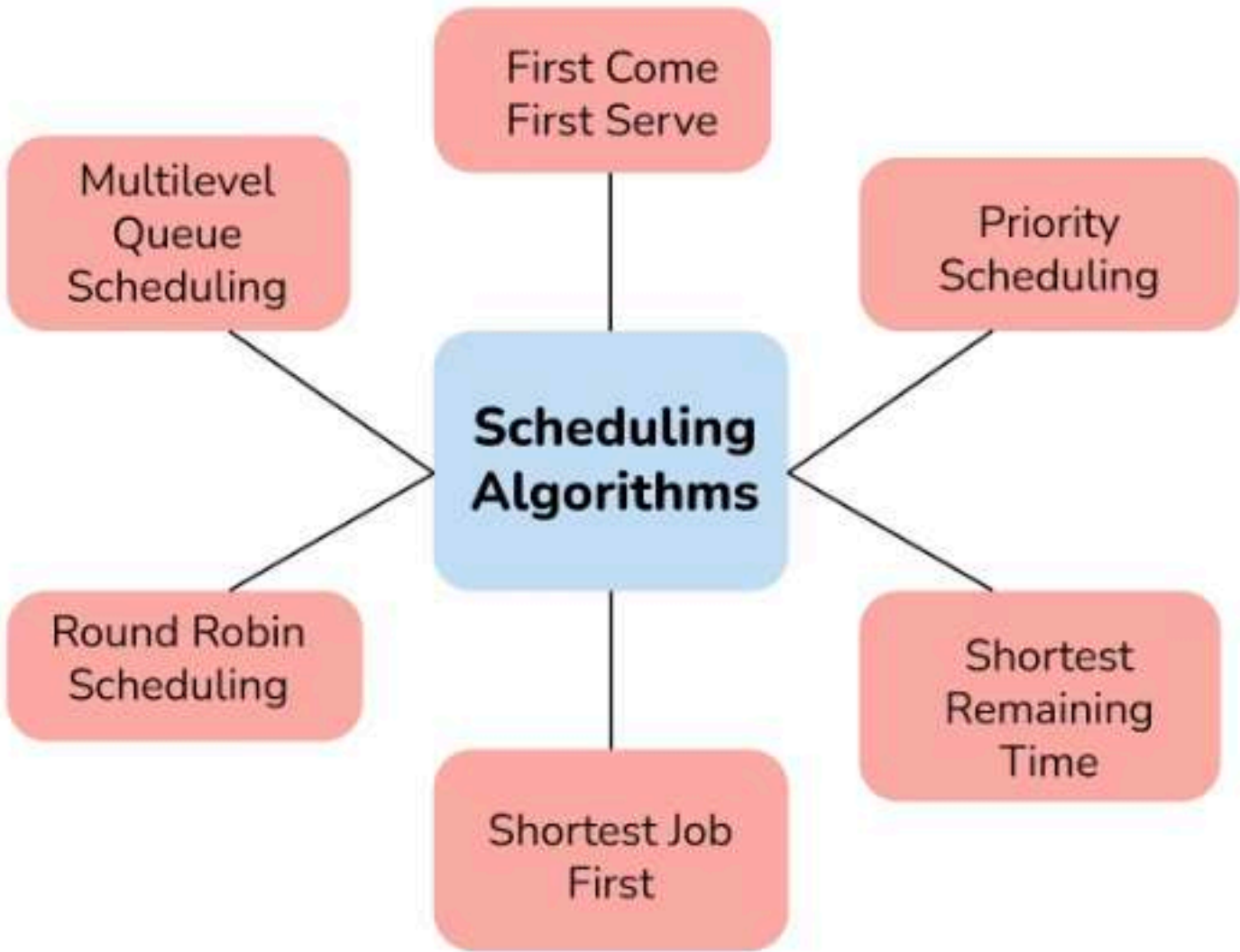
- Program code cannot change or modify itself.
- Local data for every client process needs to be stored in different disks.

30. What is a Scheduling Algorithm? Name different types of scheduling algorithms.

A scheduling algorithm is a process that is used to improve efficiency by utilizing maximum CPU and providing minimum waiting time to tasks. It simply deals with the problem of deciding which of outstanding requests is to be allocated resources. Its main aim is to reduce resource starvation and to ensure fairness amongst parties that are utilizing the resources. In simple words, it is used to allocate resources among various competing tasks.

Types of Scheduling Algorithm

There are different types of scheduling algorithms as given below:



Advanced OS Interview Questions

31. What are different types of Kernel?

There are basically five types of Kernels as given below:

- Monolithic Kernel
- MicroKernel
- Hybrid Kernel
- Nano Kernel
- Exo Kernel

32. What do you mean by Semaphore in OS? Why is it used?

[Semaphore](#) is a synchronization mechanism that is used to control access to shared resources in multi-threaded or multi-process systems. It maintains a count of available resources and provides two atomic operations: **wait()** and **signal()**. It can have a count greater than one, allowing it to control access to a finite pool of resources.

Types of Semaphores

There are two main types of semaphores:

- **Binary semaphore:** A binary semaphore is a synchronization object that can only have two values: 0 and 1. It is used to signal the availability of a single resource, such as a shared memory location or a file.
- **Counting semaphore:** A counting semaphore is a synchronization object that can have a value greater than 1. It is used to control access to a finite number of resources, such as a pool of database connections or a limited number of threads.

| Binary Semaphore                                                                                                                                                            | Mutex                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| It allows a process to signal the availability of a shared resource by setting the semaphore value to 1, and blocking other processes until the resource becomes available. | It allows mutual exclusion by ensuring that only one thread can acquire a lock on the shared resource at a time. |
| Its functions are based on signalling mechanisms.                                                                                                                           | Its func                                                                                                         |
|                                                                                                                                                                             | Get Ready with Free Mock Coding Interview                                                                        |
|                                                                                                                                                                             | Mutex c                                                                                                          |

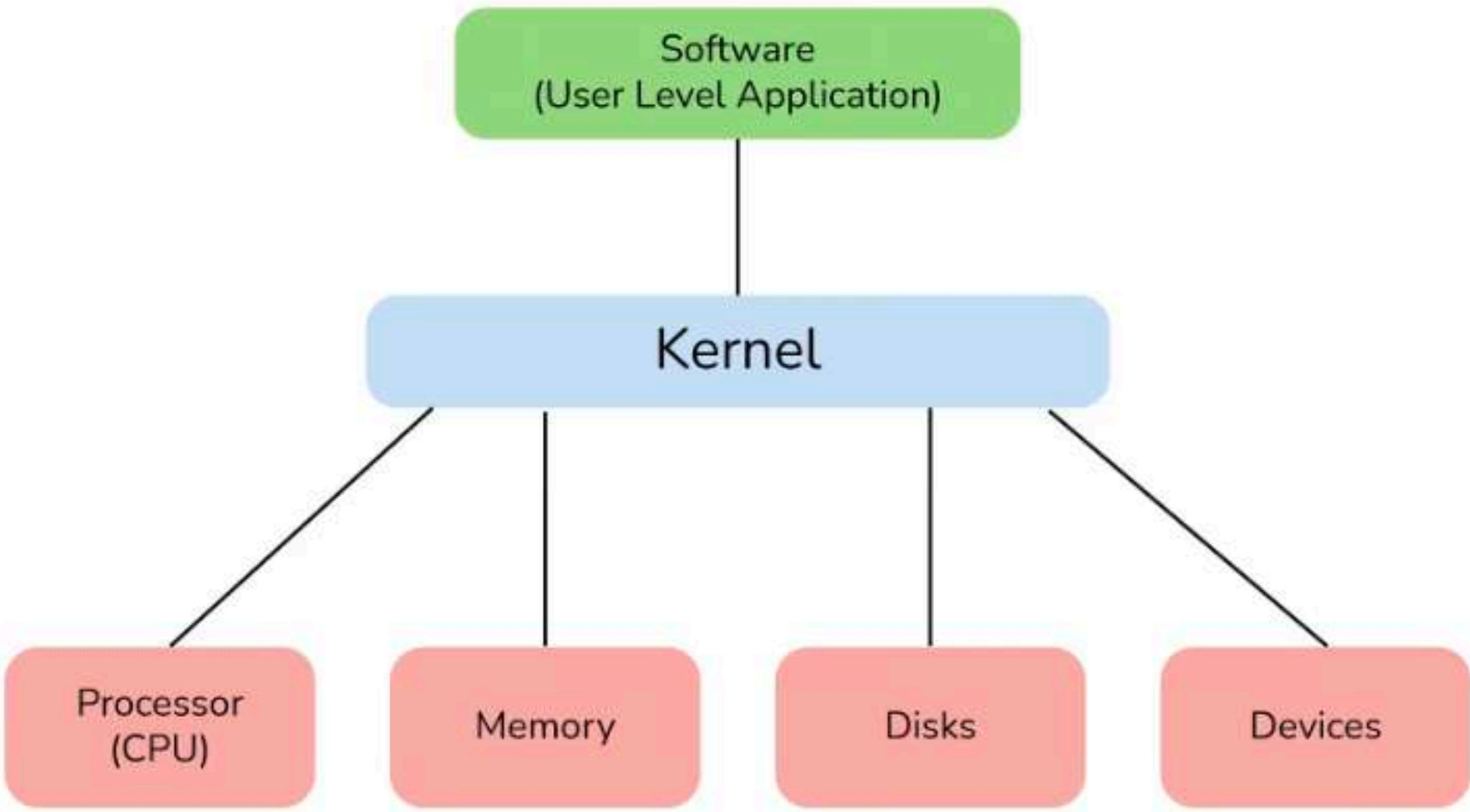


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| Binary Semaphore                                                                                                                                                               | Mutex                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| It allows a process to signal the availability of a shared resource by setting the semaphore value to 1, and blocking other processes until the resource becomes available.    | It allows mutual exclusion by ensuring that only one thread can acquire a lock on the shared resource at a time.                                                          |
| Its functions are based on signalling mechanisms.                                                                                                                              | Its functions are based upon a locking mechanism.                                                                                                                         |
| Binary semaphores can be faster than mutex in certain scenarios, such as when multiple processes need to access a shared resource and the semaphore count is greater than one. | Mutex can be slower than binary semaphore in certain scenarios, such as when the mutex has heavily contended and there is a high frequency of lock and unlock operations. |
| It is basically an integer variable that can hold either 0 or 1.                                                                                                               | It is basically an object that can hold a lock state and additional information, such as the owner of the lock and the number of threads waiting for the lock.            |

33. What is Kernel and write its main functions?

The kernel is basically a computer program usually considered as a central component or module of OS. It is responsible for handling, managing, and controlling all operations of computer systems and hardware. Whenever the system starts, the kernel is loaded first and remains in the main memory. It also acts as an interface between user applications and hardware.



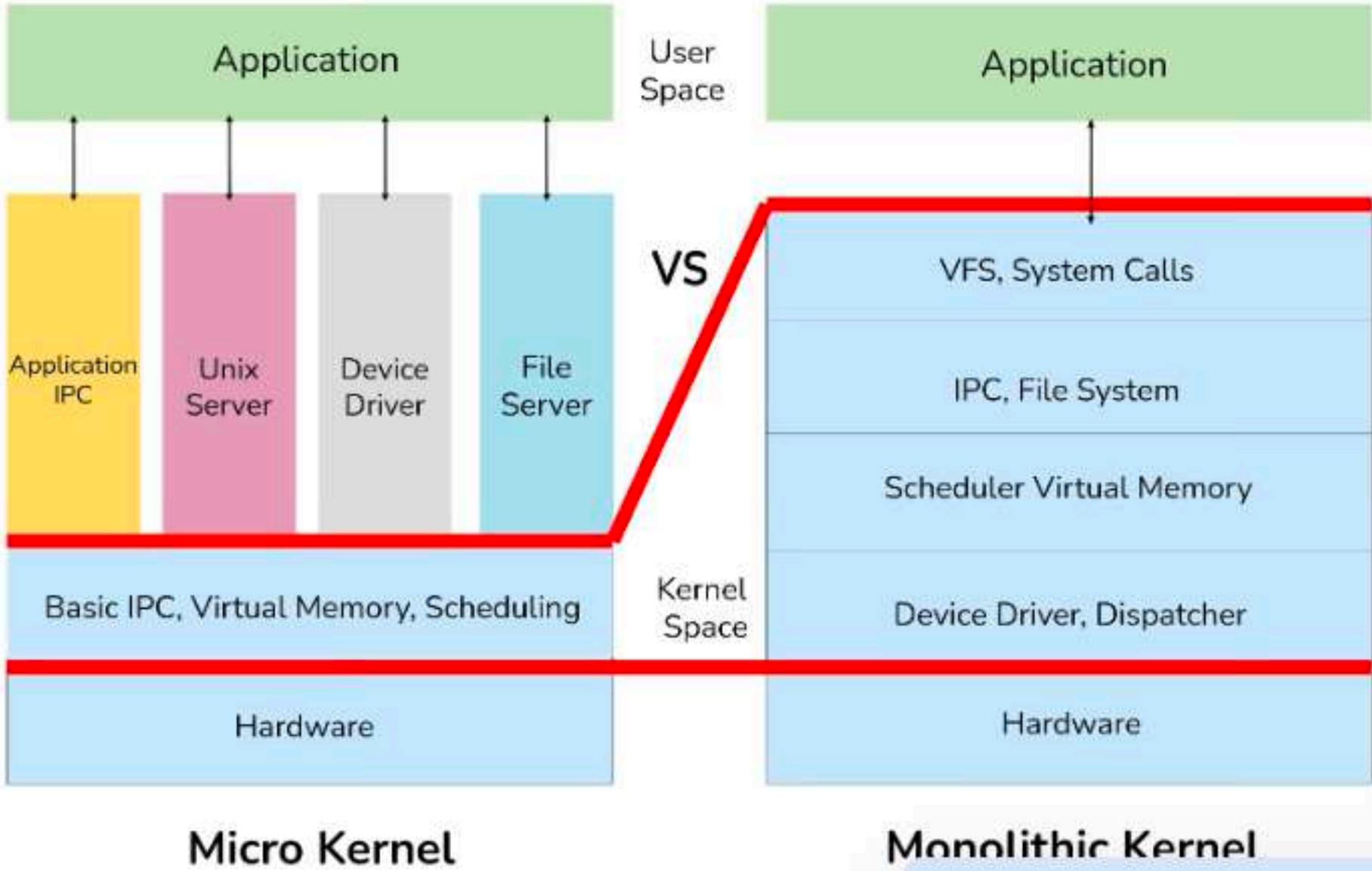
Functions of Kernel:

- It is responsible for managing all computer resources such as CPU, memory, files, processes, etc.
- It facilitates or initiates the interaction between components of hardware and software.
- It manages RAM memory so that all running processes and programs can work effectively and efficiently.
- It also controls and manages all primary tasks of the OS as well as manages access and use of various peripherals connected to the computer.
- It schedules the work done by the CPU so that the work of each user is executed as efficiently as possible.

34. Write difference between micro kernel and monolithic kernel?

**MicroKernel:** It is a minimal OS that executes only important functions of OS. It only contains a near-minimum number of features and functions that are required to implement OS.  
Example: QNX, Mac OS X, K42, etc.

**Monolithic Kernel:** It is an OS architecture that supports all basic features of computer components such as resource management, memory, file, etc.  
Example: Solaris, DOS, OpenVMS, Linux, etc.





7:56

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Space

Hardware

Hardware

Micro Kernel

Monolithic Kernel



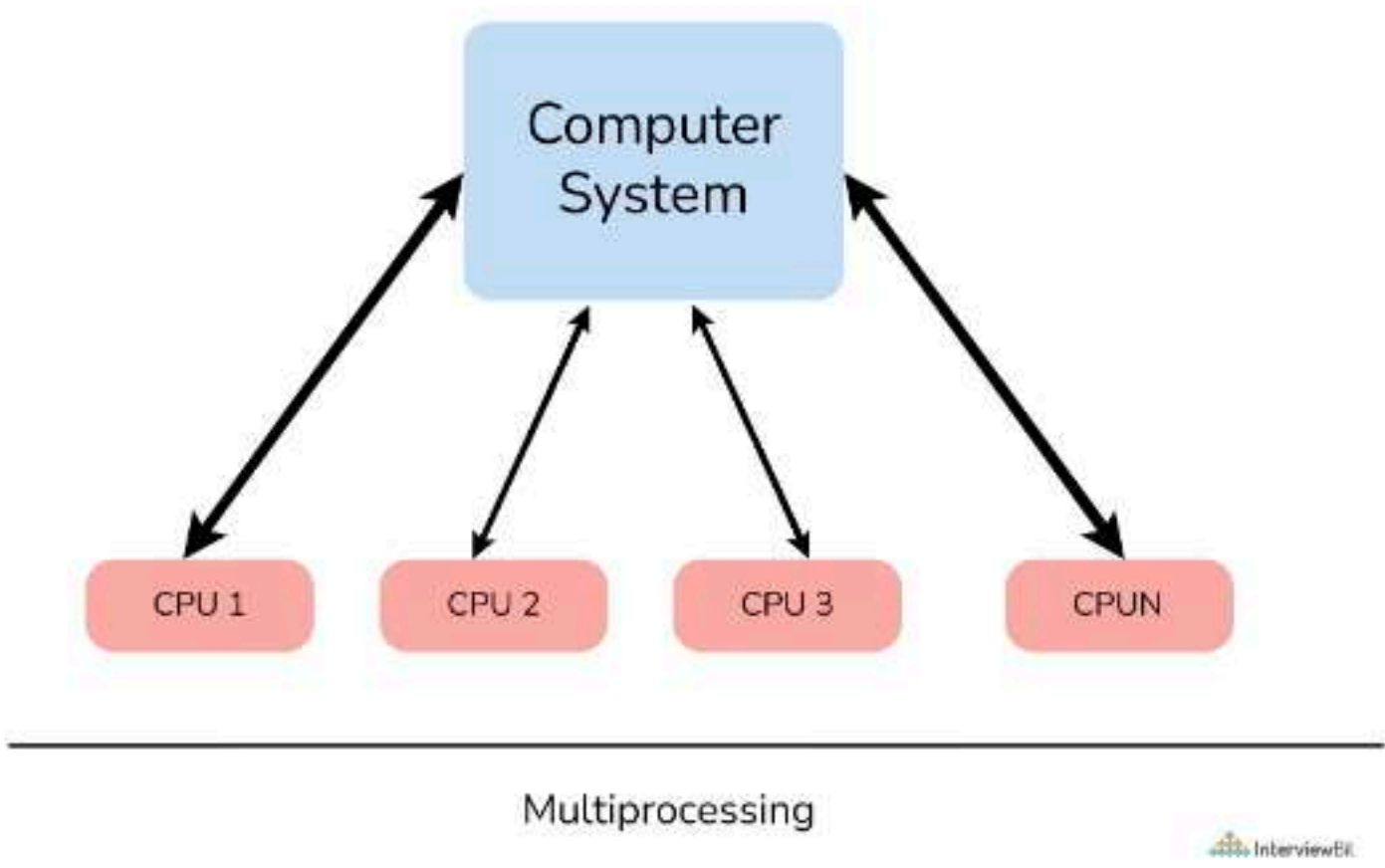
| MicroKernel                                                                                             | Monolithic Kernel                                                                                             |
|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| In this software or program, kernel services and user services are present in different address spaces. | In this software or program, kernel services and user services are usually present in the same address space. |
| It is smaller in size as compared to the monolithic kernel.                                             | It is larger in size as compared to a microkernel.                                                            |
| It is easily extendible as compared to a monolithic kernel.                                             | It is hard to as extend as compared to a microkernel.                                                         |
| If a service crashes, it does affect on working of the microkernel.                                     | If a service crashes, the whole system crashes in a monolithic kernel.                                        |
| It uses message queues to achieve inter-process communication.                                          | It uses signals and sockets to achieve inter-process communication.                                           |

35. What is SMP (Symmetric Multiprocessing)?

SMP is generally referred to as computer architecture in which the processing of programs is done by multiple processors that share a common OS and memory. SMP is very much required if you want to take advantage of multiprocessor hardware. It simply enables any processor to work on any of the tasks no matter where data or resources for that particular task are located in memory. These systems are more reliable than single-processor systems.

36. What is a time-sharing system?

It is a system that allows more than one user to access the resources of a particular system in many locations. In simple words, it performs multiple tasks on a single processor or CPU. As the name suggests, it means to share time into multiple slots in several processes. It also allows different users from different locations to use a particular computer system at the same time therefore it is considered one of the important types of OS.



```
graph TD; CS[Computer System] <--> C1[CPU 1]; CS <--> C2[CPU 2]; CS <--> C3[CPU 3]; CS <--> Cn[CPUN];
```

Multiprocessing

37. What is Context Switching?

Context switching is basically a process of saving the context of one process and loading the context of another process. It is one of the cost-effective and time-saving measures executed by CPU the because it allows multiple processes to share a single CPU. Therefore, it is considered an important part of a modern OS. This technique is used by OS to switch a process from one state to another i.e., from running state to ready state. It also allows a single CPU to handle and control various different processes or threads without even the need for additional resources.

38. What is difference between Kernel and OS?

**Kernel:** Kernel is a system program that controls all programs running on the computer. The kernel is basically a bridge between the software and hardware of the system.

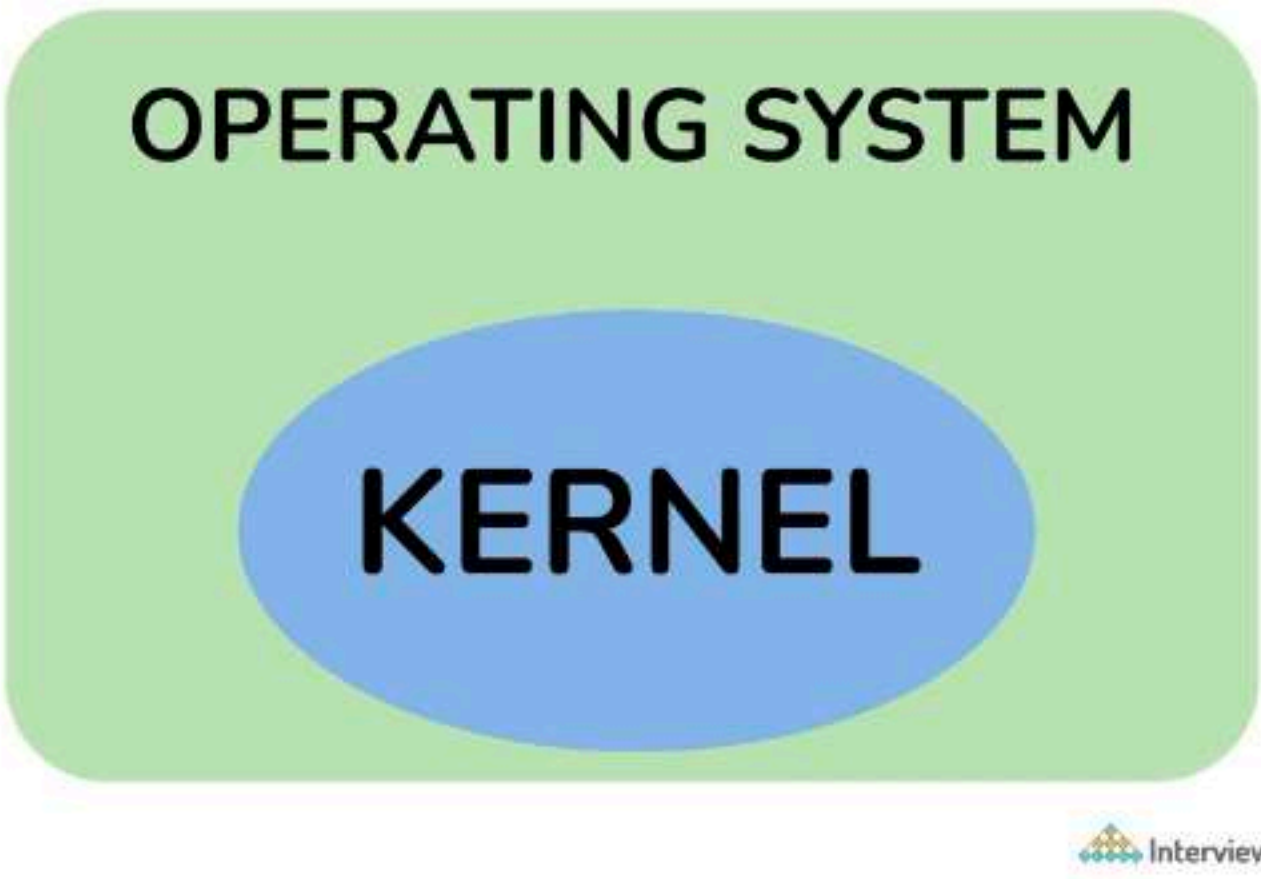
**Operating System:** Operating system is a system program that runs on the computer to provide an interface to the computer user so that they can easily operate on the computer.

OPERATING SYSTEM

KERNEL

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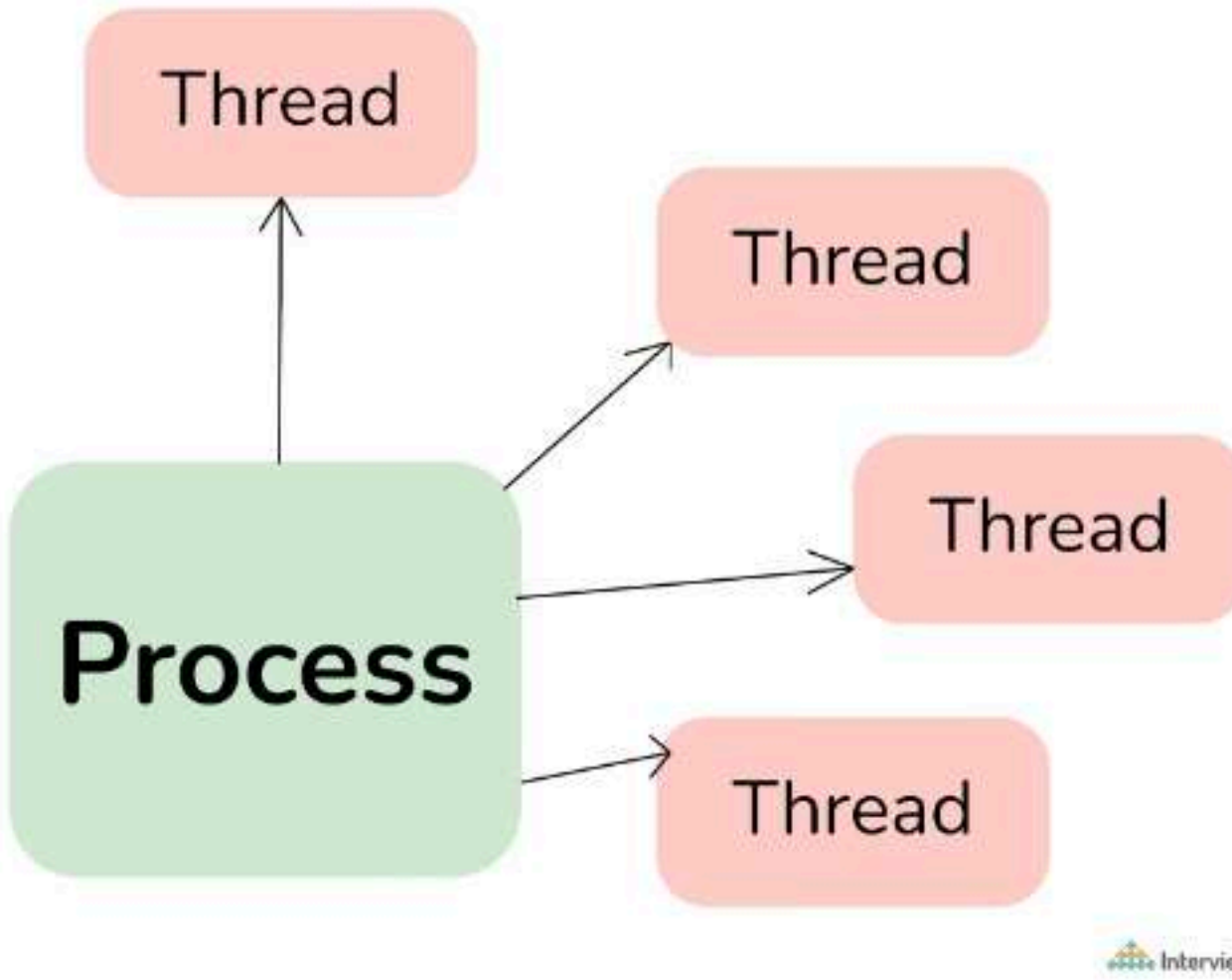


| Kernel                                                                                                          | OS                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| It is considered a central component of OS                                                                      | It is considered system software.                                                                                                                                |
| It is generally responsible for converting user commands into machine-level commands.                           | It is generally responsible for managing the resources of system.                                                                                                |
| It simply acts as an interface between hardware and applications.                                               | It simply acts as an interface between hardware and user.                                                                                                        |
| It also performs functions like process management, file management, device management, I/O communication, etc. | It also performs functions like providing security to data and files in the the system, providing access controls to users, maintaining the system privacy, etc. |
| Its type includes Microkernel, Monolithic kernel, etc.                                                          | Its type includes Single and Multiprogramming batch systems, Distributed OS, Real-time OS.                                                                       |

39. What is difference between process and thread?

**Process:** It is basically a program that is currently under execution by one or more threads. It is a very important part of the modern-day OS.

**Thread:** It is a path of execution that is composed of the program counter, thread id, stack, and set of registers within the process.



| Process                                                                       | Thread                                                                            |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| It is a computer program that is under execution.                             | It is the component or entity of the process that is the smallest execution unit. |
| These are heavy-weight operators.                                             | These are lightweight operators.                                                  |
| It has its own memory space.                                                  | It uses the memory of the process they belong to.                                 |
| It is more difficult to create a process as compared to creating a thread.    | It is easier to create a thread as compared to creating a process.                |
| It requires more resources as compared to thread.                             | It requires fewer resources as compared to processes.                             |
| It takes more time to create and terminate a process as compared to a thread. | It takes less time to create and terminate a thread as compared to a process.     |
| It usually run-in separate memory space.                                      | It usually run-in shared memory space.                                            |
| It does not share data.                                                       | It shares data with each other.                                                   |
| It can be divided into multiple threads.                                      | It can't be further subdivided.                                                   |

40. What are various sections of the process?

There are basically four sections in the process as given below:

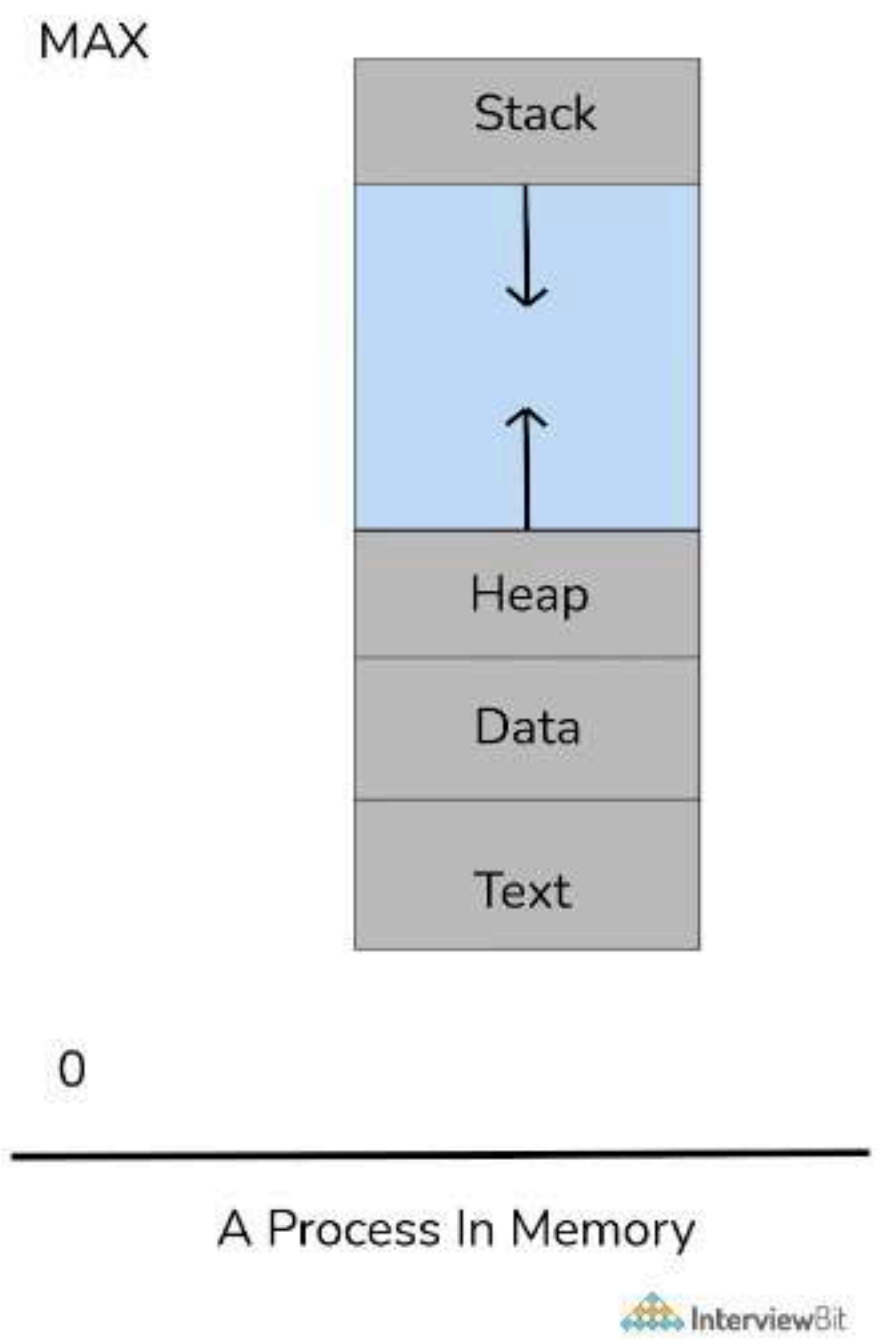


It can be divided into multiple threads.

It can't be further subdivided.

### 40. What are various sections of the process?

There are basically four sections in the process as given below:



- **Stack:** It is used for local variables and returns addresses.
- **Heap:** It is used for dynamic memory allocation.
- **Data:** It stores global and static variables.
- **Code or text:** It comprises compiled program code.

### 41. What is a deadlock in OS? What are the necessary conditions for a deadlock?

Deadlock is generally a situation where a set of processes are blocked as each process is holding resources and waits to acquire resources held by another process. In this situation, two or more processes simply try to execute simultaneously and wait for each to finish their execution because they are dependent on each other. We can see a hard problem in our system whenever a deadlock occurs in a program. It is one of the common problems you can see in multiprocessing.

#### Necessary Conditions for Deadlock

There are basically four necessary conditions for deadlock as given below:

- Mutual Exclusion
- Hold and Wait
- No Pre-emption
- Circular Wait or Resource Wait

### 42. What do you mean by Belady's Anomaly?

In the Operating System, process data is loaded in fixed-sized chunks and each chunk is referred to as a page. The processor loads these pages in the fixed-sized chunks of memory called frames. Belady's Anomaly is a phenomenon in which if we increase the number of frames in memory, then the number of page faults also increases. It is generally experienced when we use FIFO (First in First out) page replacement algorithm.

### 43. What is spooling in OS?

Spooling simply stands for Simultaneous peripheral operations online. It is referred to as putting data of various I/O jobs in a buffer. Here, buffer means a special area in memory or hard disk that can be accessible to an I/O device. It is used for mediation between a computer application and a slow peripheral. It is very useful and important because devices access or acquire data at different rates. This operation also uses disk as a very large buffer and is capable of overlapping I/O operations for one task with processor operations for another task.

#### Useful Resource

- [Free Operating System Certification Course](#)
- [Operating System MCQ](#)
- [Practice Coding](#)
- [InterviewBit Blog](#)
- [Components of Operating System](#)
- [Features of Operating System](#)
- [iOS Developer Interview Questions](#)

## Os Interview MCQs

1.Which of the following usually provides the interface to get access to the services of OS?

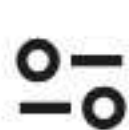
☐ System Call

☐ Library

☐ API

☐ All of the above





## Basics OS Interview Questions

### 1. What is a process and process table?

A process is an instance of a program in execution. For example, a Web Browser is a process, and a shell (or command prompt) is a process. The operating system is responsible for managing all the processes that are running on a computer and allocates each process a certain amount of time to use the processor. In addition, the [operating system](#) also allocates various other resources that processes will need, such as computer memory or disks. To keep track of the state of all the processes, the operating system maintains a table known as the process table. Inside this table, every process is listed along with the resources the process is using and the current state of the process.

### 2. What are the different states of the process?

Processes can be in one of three states: running, ready, or waiting. The running state means that the process has all the resources it needs for execution and it has been given permission by the operating system to use the processor. Only one process can be in the running state at any given time. The remaining processes are either in a waiting state (i.e., waiting for some external event to occur such as user input or disk access) or a ready state (i.e., waiting for permission to use the processor). In a real operating system, the waiting and ready states are implemented as queues that hold the processes in these states.

For more details you can refer [States of a Process in Operating Systems](#) published article.

### 3. What is a Thread?

A [thread](#) is a single sequence stream within a process. Because threads have some of the properties of processes, they are sometimes called lightweight processes. Threads are a popular way to improve the application through parallelism. For example, in a browser, multiple tabs can be different threads. MS Word uses multiple threads, one thread to format the text, another thread to process inputs, etc.

### 4. What are the differences between process and thread?

Process is program under action and thread is the smallest segment of instructions (segment of a process) that can be handled independently by a scheduler.

**Threads** are lightweight processes that share the same address space including the code section, data section and operating system resources such as the open files and signals. However, each thread has its own program counter (PC), register set and stack space allowing them to execute independently within the same process context. Unlike **processes**, threads are not fully independent entities and can communicate and synchronize more efficiently making them suitable for the concurrent and parallel execution in the multi-threaded environment.

For more details you can refer [Difference between Process and Thread](#) published article.

### 5. What are the benefits of multithreaded programming?

Multithreaded programming makes the system more responsive and enables resource sharing. It leads to the use of multiprocess architecture. It is more economical and preferred.

### 6. What is Thrashing?

Thrashing is a situation when the performance of a computer degrades or collapses. Thrashing occurs when a system spends more time processing page faults than executing transactions. While processing page faults is necessary in order to appreciate the benefits of virtual memory, thrashing has a negative effect on the system. As the page fault rate increases, more transactions need processing from the paging device. The queue at the paging device increases, resulting in increased service time for a page fault.

### 7. What is Buffer?

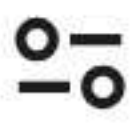
A buffer is a memory area that stores data being transferred between two devices or between a device and an application.

### 8. What is virtual memory?



Virtual memory creates an illusion that each process has one or more contiguous address spaces, each beginning at address zero. The sizes of such virtual address spaces are generally very high. The





## 8. What is virtual memory?

[Virtual memory](#) creates an illusion that each user has one or more contiguous address spaces, each beginning at address zero. The sizes of such virtual address spaces are generally very high. The idea of virtual memory is to use disk space to extend the RAM. Running processes don't need to care whether the memory is from [RAM](#) or disk. The illusion of such a large amount of memory is created by subdividing the virtual memory into smaller pieces, which can be loaded into physical memory whenever they are needed by a process.

## 9. Explain the main purpose of an operating system?

An operating system acts as an intermediary between the user of a computer and computer hardware. The purpose of an operating system is to provide an environment in which a user can execute programs conveniently and efficiently.

An operating system is a software that manages computer hardware. The hardware must provide appropriate mechanisms to ensure the correct operation of the computer system and to prevent user programs from interfering with the proper operation of the system.

## 10. What is demand paging?

The process of loading the page into memory on demand (whenever a page fault occurs) is known as [demand paging](#).

## 11. What is a kernel?

A [kernel](#) is the central component of an operating system that manages the operations of computers and hardware. It basically manages operations of memory and CPU time. It is a core component of an operating system. Kernel acts as a bridge between applications and data processing performed at the hardware level using inter-process communication and system calls.

## 12. What are the different scheduling algorithms?

1. [First-Come, First-Served \(FCFS\) Scheduling](#)
2. [Shortest-Job-Next \(SJN\) Scheduling](#)
3. Priority Scheduling
4. [Shortest Remaining Time](#)
5. [Round Robin\(RR\) Scheduling](#)
6. [Multiple-Level Queues Scheduling](#)

## 13. Describe the objective of multi-programming?

[Multi-programming](#) increases CPU utilization by organizing jobs (code and data) so that the CPU always has one to execute. The main objective of multi-programming is to keep multiple jobs in the main memory. If one job gets occupied with IO, the CPU can be assigned to other jobs.

## 14. What is the time-sharing system?

[Time-sharing](#) is a logical extension of multiprogramming. The CPU performs many tasks by switches that are so frequent that the user can interact with each program while it is running. A time-shared operating system allows multiple users to share computers simultaneously.

## 15. What problem we face in computer system without OS?

- Poor resource management
- Lack of User Interface
- No File System
- No Networking
- Error handling

## 16. Give some benefits of multithreaded programming?

A thread is also known as a lightweight process. The idea is to achieve parallelism by dividing a process into multiple threads. Threads within the same process run in shared memory space,

## 17. Briefly explain FCFS?





16. Give some benefits of multithreaded programming?

A thread is also known as a lightweight process. The idea is to achieve parallelism by dividing a process into multiple threads. Threads within the same process run in shared memory space,

17. Briefly explain FCFS?

[FCFS](#) stands for First Come First served. In the FCFS scheduling algorithm, the job that arrived first in the ready queue is allocated to the CPU and then the job that came second and so on. FCFS is a non-preemptive scheduling algorithm as a process that holds the CPU until it either terminates or performs I/O. Thus, if a longer job has been assigned to the CPU then many shorter jobs after it will have to wait.

18. What is the RR scheduling algorithm?

A round-robin scheduling algorithm is used to schedule the process fairly for each job in a time slot or quantum and interrupting the job if it is not completed by then the job comes after the other job which is arrived in the quantum time makes these scheduling fairly.

- Round-robin is cyclic in nature, so starvation doesn't occur
- Round-robin is a variant of first-come, first-served scheduling
- No priority or special importance is given to any process or task
- RR scheduling is also known as Time slicing scheduling

19. Enumerate the different RAID levels?

A redundant array of independent disks is a set of several physical disk drives that the operating system sees as a single logical unit. It played a significant role in narrowing the gap between increasingly fast processors and slow disk drives. RAID has different levels:

- Level-0
- Level-1
- Level-2
- Level-3
- Level-4
- Level-5
- Level-6

20. What is Banker's algorithm?

The [banker's algorithm](#) is a resource allocation and deadlock avoidance algorithm that tests for safety by simulating the allocation for the predetermined maximum possible amounts of all resources, then makes an "s-state" check to test for possible activities, before deciding whether allocation should be allowed to continue.

21. State the main difference between logical and physical address space?

| Parameter     | LOGICAL ADDRESS                                                                                         | PHYSICAL ADDRESS                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Basic         | Logical address is generated by the CPU.                                                                | It is located in a memory unit.                                                                    |
| Address Space | Logical Address Space is a set of all logical addresses generated by the CPU in reference to a program. | Physical Address is a set of all physical addresses mapped to the corresponding logical addresses. |
| Visibility    | Users can view the logical address of a program.                                                        | Users can never view the physical address of the program.                                          |
| Generation    | generated by the CPU.                                                                                   | Computed by MMU.                                                                                   |
| Access        | The user can use the logical address to access the physical address.                                    | The user can indirectly access physical addresses but not directly.                                |

22. How does dynamic loading aid in better memory space utilization?

With dynamic loading, a routine is not loaded until it is called. This method is especially useful when large amounts of code are needed in order to handle infrequently occurring cases such as error routines.

23. What are overlays?

The concept of overlays is that whenever a program is running it will not use the complete program at the same time, it will use only some part of it. Then overlay concept says that whatever part you



### 23. What are overlays?

The concept of overlays is that whenever a process is running it will not use the complete program at the same time, it will use only some part of it. Then overlay concept says that whatever part you required, you load it and once the part is done, then you just unload it, which means just pull it back and get the new part you required and run it. Formally, "The process of transferring a block of program code or other data into internal memory, replacing what is already stored".

### 24. What is fragmentation?

Processes are stored and removed from memory, which makes free memory space, which is too little to even consider utilizing by different processes. Suppose, that process is not ready to dispense to memory blocks since its little size and memory hinder consistently staying unused is called [fragmentation](#). This kind of issue occurs during a dynamic memory allotment framework when free blocks are small, so it can't satisfy any request.

### 25. What is the basic function of paging?

[Paging](#) is a method or technique which is used for non-contiguous memory allocation. It is a fixed-size partitioning theme (scheme). In paging, both main memory and secondary memory are divided into equal fixed-size partitions. The partitions of the secondary memory area unit and the main memory area unit are known as pages and frames respectively.

Paging is a memory management method accustomed fetch processes from the secondary memory into the main memory in the form of pages. in paging, each process is split into parts wherever the size of every part is the same as the page size. The size of the last half could also be but the page size. The pages of the process area unit hold on within the frames of main memory relying upon their accessibility

### 26. How does swapping result in better memory management?

[Swapping](#) is a simple memory/process management technique used by the operating system(os) to increase the utilization of the processor by moving some blocked processes from the main memory to the secondary memory thus forming a queue of the temporarily suspended processes and the execution continues with the newly arrived process. During regular intervals that are set by the operating system, processes can be copied from the main memory to a backing store and then copied back later. Swapping allows more processes to be run that can fit into memory at one time

### 27. Write a name of classic synchronization problems?

- Bounded-buffer
- [Readers-writers](#)
- [Dining.philosophers](#)
- Sleeping barber

### 28. What is the Direct Access Method?

The direct Access method is based on a disk model of a file, such that it is viewed as a numbered sequence of blocks or records. It allows arbitrary blocks to be read or written. Direct access is advantageous when accessing large amounts of information. Direct memory access (DMA) is a method that allows an input/output (I/O) device to send or receive data directly to or from the main memory, bypassing the CPU to speed up memory operations. The process is managed by a chip known as a DMA controller ([DMAC](#)).

### 29. When does thrashing occur?

Thrashing occurs when processes on the system frequently access pages, not available memory.

### 30. What is the best page size when designing an operating system?

The best paging size varies from system to system, so there is no single best when it comes to page size. There are different factors to consider in order to come up with a suitable page size, such as page table, paging time, and its effect on the overall efficiency of the operating system.

### 31. What is multitasking?

[Multitasking](#) is a logical extension of a multiprogramming system that supports multiple programs to run concurrently. In multitasking, more than one task is executed at the same time. In this technique, the multiple tasks, also known as processes, share common processing resources such as a CPU.

### 32. What is caching?



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### 32. What is caching?

The cache is a smaller and faster memory that stores copies of the data from frequently used main memory locations. There are various different independent caches in a CPU, which store instructions and data. Cache memory is used to reduce the average time to access data from the Main memory.

### 33. What is spooling?

[Spooling](#) refers to simultaneous peripheral operations online, spooling refers to putting jobs in a buffer, a special area in memory, or on a disk where a device can access them when it is ready. Spooling is useful because devices access data at different rates.

### 34. What is the functionality of an Assembler?

The [Assembler](#) is used to translate the program written in Assembly language into machine code. The source program is an input of an assembler that contains assembly language instructions. The output generated by the assembler is the object code or machine code understandable by the computer.

### 35. What are interrupts?

The interrupts are a signal emitted by hardware or software when a process or an event needs immediate attention. It alerts the processor to a high-priority process requiring interruption of the current working process. In I/O devices one of the bus control lines is dedicated to this purpose and is called the Interrupt Service Routine (ISR).

### 36. What is GUI?

GUI is short for [Graphical User Interface](#). It provides users with an interface wherein actions can be performed by interacting with icons and graphical symbols.

### 37. What is preemptive multitasking?

Preemptive multitasking is a type of multitasking that allows computer programs to share operating systems (OS) and underlying hardware resources. It divides the overall operating and computing time between processes, and the switching of resources between different processes occurs through predefined criteria.

### 38. What is a pipe and when is it used?

A Pipe is a technique used for [inter-process communication](#). A pipe is a mechanism by which the output of one process is directed into the input of another process. Thus it provides a one-way flow of data between two related processes.

### 39. What are the advantages of semaphores?

- They are machine-independent.
- Easy to implement.
- Correctness is easy to determine.
- Can have many different critical sections with different semaphores.
- Semaphores acquire many resources simultaneously.
- No waste of resources due to busy waiting.

### 40. What is a bootstrap program in the OS?

Bootstrapping is the process of loading a set of instructions when a computer is first turned on or booted. During the startup process, diagnostic tests are performed, such as the power-on self-test (POST), which set or checks configurations for devices and implements routine testing for the connection of peripherals, hardware, and external memory devices. The bootloader or bootstrap program is then loaded to initialize the OS.

### 41. What is IPC?



program is then loaded to initialize the OS.

#### 41. What is IPC?

Inter-process communication (IPC) is a mechanism that allows processes to communicate with each other and synchronize their actions. The communication between these processes can be seen as a method of cooperation between them.

#### 42. What are the different IPC mechanisms?

These are the methods in IPC:

- **Pipes (Same Process):** This allows a flow of data in one direction only. Analogous to simplex systems (Keyboard). Data from the output is usually buffered until the input process receives it which must have a common origin.
- **Named Pipes (Different Processes):** This is a pipe with a specific name it can be used in processes that don't have a shared common process origin. E.g. FIFO where the details written to a pipe are first named.
- **Message Queuing:** This allows messages to be passed between processes using either a single queue or several message queues. This is managed by the system kernel these messages are coordinated using an API.
- **Semaphores:** This is used in solving problems associated with synchronization and avoiding race conditions. These are integer values that are greater than or equal to 0.
- **Shared Memory:** This allows the interchange of data through a defined area of memory. Semaphore values have to be obtained before data can get access to shared memory.
- **Sockets:** This method is mostly used to communicate over a network between a client and a server. It allows for a standard connection which is computer and OS independent

#### 43. What is the difference between preemptive and non-preemptive scheduling?

- In preemptive scheduling, the CPU is allocated to the processes for a limited time whereas, in Non-preemptive scheduling, the CPU is allocated to the process till it terminates or switches to waiting for the state.
- The executing process in preemptive scheduling is interrupted in the middle of execution when a higher priority one comes whereas, the executing process in non-preemptive scheduling is not interrupted in the middle of execution and waits till its execution.
- In Preemptive Scheduling, there is the overhead of switching the process from the ready state to the running state, vice-verse, and maintaining the ready queue. Whereas the case of non-preemptive scheduling has no overhead of switching the process from running state to ready state.
- In preemptive scheduling, if a high-priority process frequently arrives in the ready queue then the process with low priority has to wait for a long, and it may have to starve. On the other hand, in non-preemptive scheduling, if CPU is allocated to the process having a larger burst time then the processes with a small burst time may have to starve.
- Preemptive scheduling attains flexibility by allowing the critical processes to access the CPU as they arrive in the ready queue, no matter what process is executing currently. Non-preemptive scheduling is called rigid as even if a critical process enters the ready queue the process running CPU is not disturbed.
- Preemptive Scheduling has to maintain the integrity of shared data that's why it is cost associative which is not the case with Non-preemptive Scheduling.

#### 44. What is the zombie process?

A process that has finished the execution but still has an entry in the process table to report to its parent process is known as a zombie process. A child process always first becomes a zombie before being removed from the process table. The parent process reads the exit status of the child process which reaps off the child process entry from the process table.

#### 45. What are orphan processes?

A process whose parent process no more exists i.e. either finished or terminated without waiting for its child process to terminate is called an orphan process.

#### 46. What are starvation and aging in OS?

**Starvation:** Starvation is a resource management problem where a process does not get the resources it needs for a long time because the resources are being allocated to other processes.

**Aging:** Aging is a technique to avoid starvation in a scheduling system. It works by adding an aging factor to the priority of each request. The aging factor must increase the priority of the request as time passes and must ensure that a request will eventually be the highest priority request

#### 47. Write about monolithic kernel?









50. What is the difference between process and thread?

| Process                                                                          | Thread                                                                                              |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Process means any program is in execution.                                       | Thread means a segment of a process.                                                                |
| The process is less efficient in terms of communication.                         | Thread is more efficient in terms of communication.                                                 |
| The process is isolated.                                                         | Threads share memory.                                                                               |
| The process is called heavyweight the process.                                   | Thread is called lightweight process.                                                               |
| Process switching uses, another process interface in operating system.           | Thread switching does not require to call an operating system and cause an interrupt to the kernel. |
| If one process is blocked then it will not affect the execution of other process | The second, thread in the same task could not run, while one server thread is blocked.              |
| The process has its own Process Control Block, Stack and Address Space.          | Thread has Parents' PCB, its own Thread Control Block and Stack and common Address space.           |

51. What is PCB?

the process control block (PCB) is a block that is used to track the process's execution status. A process control block (PCB) contains information about the process, i.e. registers, quantum, priority, etc. The process table is an array of PCBs, that means logically contains a PCB for all of the current processes in the system.

52. When is a system in a safe state?

The set of dispatchable processes is in a safe state if there exists at least one temporal order in which all processes can be run to completion without resulting in a deadlock.

53. What is Cycle Stealing?

cycle stealing is a method of accessing computer memory (RAM) or bus without interfering with the CPU. It is similar to direct memory access (DMA) for allowing I/O controllers to read or write RAM without CPU intervention.

54. What are a Trap and Trapdoor?

A trap is a software interrupt, usually the result of an error condition, and is also a non-maskable interrupt and has the highest priority Trapdoor is a secret undocumented entry point into a program used to grant access without normal methods of access authentication.

55. Write a difference between program and process?

| Program                                                                                                     | Process                                                                                                            |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Program contains a set of instructions designed to complete a specific task.                                | Process is an instance of an executing program.                                                                    |
| Program is a passive entity as it resides in the secondary memory.                                          | Process is an active entity as it is created during execution and loaded into the main memory.                     |
| The program exists in a single place and continues to exist until it is deleted.                            | Process exists for a limited span of time as it gets terminated after the completion of a task.                    |
| A program is a static entity.                                                                               | The process is a dynamic entity.                                                                                   |
| Program does not have any resource requirement, it only requires memory space for storing the instructions. | Process has a high resource requirement, it needs resources like CPU, memory address, and I/O during its lifetime. |
| The program does not have any control block.                                                                | The process has its own control block called Process Control Block.                                                |

56. What is a dispatcher?



55. Write a difference between program and process?

| Program                                                                                                     | Process                                                                                                            |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Program contains a set of instructions designed to complete a specific task.                                | Process is an instance of an executing program.                                                                    |
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56. What is a dispatcher?

The dispatcher is the module that gives process control over the CPU after it has been selected by the short-term scheduler. This function involves the following:

- Switching context
- Switching to user mode
- Jumping to the proper location in the user program to restart that program

57. Define the term dispatch latency?

Dispatch latency can be described as the amount of time it takes for a system to respond to a request for a process to begin operation. With a scheduler written specifically to honor application priorities, real-time applications can be developed with a bounded dispatch latency.

58. What are the goals of CPU scheduling?

- Max CPU utilization [Keep CPU as busy as possible]
- Fair allocation of CPU.
- Max throughput [Number of processes that complete their execution per time unit]
- Min turnaround time [Time taken by a process to finish execution]
- Min waiting time [Time a process waits in ready queue]
- Min response time [Time when a process produces the first response]

59. What is a critical- section?

When more than one processes access the same code segment that segment is known as the critical section. The critical section contains shared variables or resources which are needed to be synchronized to maintain the consistency of data variables. In simple terms, a critical section is a group of instructions/statements or regions of code that need to be executed atomically such as accessing a resource (file, input or output port, global data, etc.).

60. Write the name of synchronization techniques?

- Mutexes
- Condition variables
- Semaphores
- File locks





Intermediate OS Interview Questions

61. Write a difference between a user-level thread and a kernel-level thread?

| User-level thread                                                                           | Kernel level thread                                                                               |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| User threads are implemented by users.                                                      | kernel threads are implemented by OS.                                                             |
| OS doesn't recognize user-level threads.                                                    | Kernel threads are recognized by OS.                                                              |
| Implementation of User threads is easy.                                                     | Implementation of the perform kernel thread is complicated.                                       |
| Context switch time is less.                                                                | Context switch time is more.                                                                      |
| Context switch requires no hardware support.                                                | Hardware support is needed.                                                                       |
| If one user-level thread performs a blocking operation then entire process will be blocked. | If one kernel thread perform a the blocking operation then another thread can continue execution. |
| User-level threads are designed as dependent threads.                                       | Kernel level threads are designed as independent threads.                                         |

62. Write down the advantages of multithreading?

Some of the most important benefits of MT are:

- Improved throughput. Many concurrent compute operations and I/O requests within a single process.
- Simultaneous and fully symmetric use of multiple processors for computation and I/O.
- Superior application responsiveness. If a request can be launched on its own thread, applications do not freeze or show the "hourglass". An entire application will not block or otherwise wait, pending the completion of another request.
- Improved server responsiveness. Large or complex requests or slow clients don't block other requests for service. The overall throughput of the server is much greater.
- Minimized system resource usage. Threads impose minimal impact on system resources. Threads require less overhead to create, maintain, and manage than a traditional process.
- Program structure simplification. Threads can be used to simplify the structure of complex applications, such as server-class and multimedia applications. Simple routines can be written for each activity, making complex programs easier to design and code, and more adaptive to a wide variation in user demands.
- Better communication. Thread synchronization functions can be used to provide enhanced process-to-process communication. In addition, sharing large amounts of data through separate threads of execution within the same address space provides extremely high-bandwidth, low-latency communication between separate tasks within an application

63. Difference between Multithreading and Multitasking?

| Multi-threading                                                                               | Multi-tasking                                                                               |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Multiple threads are executing at the same time at the same or different part of the program. | Several programs are executed concurrently.                                                 |
| CPU switches between multiple threads.                                                        | CPU switches between multiple tasks and processes.                                          |
| It is the process of a lightweight part.                                                      | It is a heavyweight process.                                                                |
| It is a feature of the process.                                                               | It is a feature of the OS.                                                                  |
| Multi-threading is sharing of computing resources among threads of a single process.          | Multitasking is sharing of computing resources(CPU, memory, devices, etc.) among processes. |

64. What are the drawbacks of semaphores?

- Priority Inversion is a big limitation of semaphores.
- Their use is not enforced but is by convention only.
- The programmer has to keep track of all calls wait and signal the semaphore.
- With improper use, a process may block Such a situation is called Deadlock.



resources among threads of a single process.

processes.

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- With improper use, a process may block indefinitely. Such a situation is called Deadlock.

## 65. What is Peterson's approach?

It is a concurrent programming algorithm. It is used to synchronize two processes that maintain the mutual exclusion for the shared resource. It uses two variables, a bool array flag of size 2 and an int variable turn to accomplish it.

## 66. Define the term Bounded waiting?

A system is said to follow bounded waiting conditions if a process wants to enter into a critical section will enter in some finite time.

## 67. What are the solutions to the critical section problem?

There are three solutions to the critical section problem:

- Software solutions
- Hardware solutions
- Semaphores

## 68. What is a Banker's algorithm?

The banker's algorithm is a resource allocation and deadlock avoidance algorithm that tests for safety by simulating the allocation for the predetermined maximum possible amounts of all resources, then makes an "s-state" check to test for possible activities, before deciding whether allocation should be allowed to continue.

## 69. What is concurrency?

A state in which a process exists simultaneously with another process than those it is said to be concurrent.

## 70. Write a drawback of concurrency?

- It is required to protect multiple applications from one another.
- It is required to coordinate multiple applications through additional mechanisms.
- Additional performance overheads and complexities in operating systems are required for switching among applications.
- Sometimes running too many applications concurrently leads to severely degraded performance.

## 71. What are the necessary conditions which can lead to a deadlock in a system?

**Mutual Exclusion:** There is a resource that cannot be shared.

**Hold and Wait:** A process is holding at least one resource and waiting for another resource, which is with some other process.

**No Preemption:** The operating system is not allowed to take a resource back from a process until the process gives it back.

**Circular Wait:** A set of processes waiting for each other in circular form.

## 72. What are the issues related to concurrency?

- **Non-atomic:** Operations that are non-atomic but interruptible by multiple processes can cause problems.
- **Race conditions:** A race condition occurs if the outcome depends on which of several processes gets to a point first.
- **Blocking:** Processes can block waiting for resources. A process could be blocked for a long period of time waiting for input from a terminal. If the process is required to periodically update some data, this would be very undesirable.
- **Starvation:** It occurs when a process does not obtain service to progress.
- **Deadlock:** It occurs when two processes are blocked and hence neither can proceed to execute.

## 73. Why do we use precedence graphs?

A precedence graph is a directed acyclic graph that is used to show the execution level of several processes in the operating system. It has the following properties also:



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- [Starvation](#): It occurs when a process does not obtain service to progress.
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73. Why do we use precedence graphs?

A precedence graph is a directed acyclic graph that is used to show the execution level of several processes in the operating system. It has the following properties also:

- Nodes of graphs correspond to individual statements of program code.
- An edge between two nodes represents the execution order.
- A directed edge from node A to node B shows that statement A executes first and then Statement B executes

74. Explain the resource allocation graph?

The resource allocation graph is explained to us what is the state of the system in terms of processes and resources. One of the advantages of having a diagram is, sometimes it is possible to see a deadlock directly by using RAG.

75. What is a deadlock?

Deadlock is a situation when two or more processes wait for each other to finish and none of them ever finish. Consider an example when two trains are coming toward each other on the same track and there is only one track, none of the trains can move once they are in front of each other. A similar situation occurs in operating systems when there are two or more processes that hold some resources and wait for resources held by other(s).

76. What is the goal and functionality of memory management?

The goal and functionality of memory management are as follows;

- Relocation
- Protection
- Sharing
- Logical organization
- Physical organization

77. Write a difference between physical address and logical address?

| Parameters | Logical address                                                                                                      | Physical Address                                                                                                  |
|------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Basic      | It is the virtual address generated by CPU.                                                                          | The physical address is a location in a memory unit.                                                              |
| Address    | Set of all logical addresses generated by the CPU in reference to a program is referred to as Logical Address Space. | Set of all physical addresses mapped to the corresponding logical addresses is referred to as a Physical Address. |
| Visibility | The user can view the logical address of a program.                                                                  | The user can never view the physical address of the program                                                       |
| Access     | The user uses the logical address to access the physical address.                                                    | The user can not directly access the physical address                                                             |
| Generation | The Logical Address is generated by the CPU                                                                          | Physical Address is Computed by MMU                                                                               |

78. Explain address binding?

The Association of program instruction and data to the actual physical memory locations is called Address Binding.

79. Write different types of address binding?

Address Binding is divided into three types as follows.

- Compile-time Address Binding
- [Load time Address Binding](#)
- [Execution time Address Binding](#)

80. Write an advantage of dynamic allocation algorithms?

- When we do not know how much amount of memory would be needed for the program beforehand.







- Define Compaction.

84. Write a difference between paging and segmentation?

| Paging                                                                            | Segmentation                                                                                              |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| In paging, program is divided into fixed or mounted-size pages.                   | In segmentation, the program is divided into variable-size sections.                                      |
| For the paging operating system is accountable.                                   | For segmentation compiler is accountable.                                                                 |
| Page size is determined by hardware.                                              | Here, the section size is given by the user.                                                              |
| It is faster in comparison of segmentation.                                       | Segmentation is slow.                                                                                     |
| Paging could result in internal fragmentation.                                    | Segmentation could result in external fragmentation.                                                      |
| In paging, logical address is split into that page number and page offset.        | Here, logical address is split into section number and section offset.                                    |
| Paging comprises a page table which encloses the base address of every page.      | While segmentation also comprises the segment table which encloses the segment number and segment offset. |
| A page table is employed to keep up the page data.                                | Section Table maintains the section data.                                                                 |
| In paging, operating system must maintain a free frame list.                      | In segmentation, the operating system maintains a list of holes in the main memory.                       |
| Paging is invisible to the user.                                                  | Segmentation is visible to the user.                                                                      |
| In paging, processor needs page number, offset to calculate the absolute address. | In segmentation, the processor uses segment number, and offset to calculate the full address.             |

85. Write a definition of Associative Memory and Cache Memory?

| Associative Memory                                                      | Cache Memory                                                       |
|-------------------------------------------------------------------------|--------------------------------------------------------------------|
| A memory unit accessed by content is called associative memory.         | Fast and small memory is called cache memory.                      |
| It reduces the time required to find the item stored in memory.         | It reduces the average memory access time.                         |
| Here data is accessed by its content.                                   | Here, data are accessed by their address.                          |
| It is used where search time is very short.                             | It is used when a particular group of data is accessed repeatedly. |
| Its basic characteristic is its logic circuit for matching its content. | Its basic characteristic is its fast access                        |

86. What is "Locality of reference"?

The locality of reference refers to a phenomenon in which a computer program tends to access the same set of memory locations for a particular time period. In other words, Locality of Reference refers to the tendency of the computer program to access instructions whose addresses are near one another.

87. Write down the advantages of virtual memory?

- A higher degree of multiprogramming.
- Allocating memory is easy and cheap
- Eliminates external fragmentation
- Data (page frames) can be scattered all over the PM
- Pages are mapped appropriately anyway
- Large programs can be written, as the virtual space available is huge compared to physical memory.
- Less I/O required leads to faster and easy running of processes.



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- Allocating memory is easy and cheap
- Eliminates external fragmentation
- Data (page frames) can be scattered all over the PM
- Pages are mapped appropriately anyway
- Large programs can be written, as the virtual space available is huge compared to physical memory.
- Less I/O required leads to faster and easy swapping of processes.
- More physical memory is available, as programs are stored on virtual memory, so they occupy very less space on actual physical memory.
- More efficient swapping

## 88. How to calculate performance in virtual memory?

The performance of virtual memory of a virtual memory management system depends on the total number of page faults, which depend on "paging policies" and "frame allocation".

Effective access time =  $(1-p) \times \text{Memory access time} + p \times \text{page fault time}$

## 89. Write down the basic concept of the file system?

A file is a collection of related information that is recorded on secondary storage. Or file is a collection of logically related entities. From the user's perspective, a file is the smallest allotment of logical secondary storage.

## 90. Write the names of different operations on file?

Operation on file:

- Create
- Open
- Read
- Write
- Rename
- Delete
- Append
- Truncate
- Close

## 91. Define the term Bit-Vector?

A Bitmap or Bit Vector is a series or collection of bits where each bit corresponds to a disk block. The bit can take two values: 0 and 1: 0 indicates that the block is allocated and 1 indicates a free block.

## 92. What is a File allocation table?

FAT stands for File Allocation Table and this is called so because it allocates different files and folders using tables. This was originally designed to handle small file systems and disks. A file allocation table (FAT) is a table that an operating system maintains on a hard disk that provides a map of the cluster (the basic units of logical storage on a hard disk) that a file has been stored in.

## 93. What is rotational latency?

Rotational Latency: Rotational Latency is the time taken by the desired sector of the disk to rotate into a position so that it can access the read/write heads. So the disk scheduling algorithm that gives minimum rotational latency is better.

## 94. What is seek time?

Seek Time: Seek time is the time taken to locate the disk arm to a specified track where the data is to be read or written. So the disk scheduling algorithm that gives a minimum average seek time is better.



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## Advanced OS Interview Questions

### 95. What is Belady's Anomaly?

Bélády's anomaly is an anomaly with some page replacement policies increasing the number of page frames resulting in an increase in the number of page faults. It occurs when the First In First Out page replacement is used.

### 96. What happens if a non-recursive mutex is locked more than once?

Deadlock. If a thread that had already locked a mutex, tries to lock the mutex again, it will enter into the waiting list of that mutex, which results in a deadlock. It is because no other thread can unlock the mutex. An operating system implementer can exercise care in identifying the owner of the mutex and return it if it is already locked by the same thread to prevent deadlocks.

### 97. What are the advantages of a multiprocessor system?

There are some main advantages of a multiprocessor system:

- Enhanced performance.
- Multiple applications.
- Multi-tasking inside an application.
- High throughput and responsiveness.
- Hardware sharing among CPUs.

### 98. What are real-time systems?

A real-time system means that the system is subjected to real-time, i.e., the response should be guaranteed within a specified timing constraint or the system should meet the specified deadline.

### 99. How to recover from a deadlock?

We can recover from a deadlock by following methods:

- Process termination
  - Abort all the deadlock processes
  - Abort one process at a time until the deadlock is eliminated
- Resource preemption
  - Rollback
  - Selecting a victim

### 100. What factors determine whether a detection algorithm must be utilized in a deadlock avoidance system?

One is that it depends on how often a deadlock is likely to occur under the implementation of this algorithm. The other has to do with how many processes will be affected by deadlock when this algorithm is applied.

### 101. Explain the resource allocation graph?

The resource allocation graph is explained to us what is the state of the system in terms of processes and resources. One of the advantages of having a diagram is, sometimes it is possible to see a deadlock directly by using RAG.

**Also check:** [Last Minute Notes – Operating Systems](#)

We will soon be covering more Operating System questions.

## Conclusion

In conclusion, the field of operating systems is a crucial aspect of computer science, and a thorough understanding of its concepts is essential for anyone looking to excel in this area. By reviewing the

Top 2024 100+ operating systems interview questions we have compiled, you can gain a deeper understanding of the key principles and concepts of OS and be better prepared to tackle any